

4. Signal Representation

Internet of Things

Executive Summary

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Motivation: Understanding signals is fundamental for IoT systems, as they enable the interaction between the physical world and digital processing.

Problem: It is necessary to model, represent, and process signals, as well as convert them from analog to digital for efficient computation and communication.

Overview:

- Introduction to signals, types, and transducers.
- Signal representation in time and frequency domains.
- Sampling and quantization processes.
- Introduction to the Fourier Transform.
- Hands-on lab: capacitors and low-pass filters.

Conclusion: Signal processing provides the foundation for IoT systems by enabling the acquisition, transformation, and filtering of real-world data.

Outline

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Introduction

Signals and sensors

Signal representation

Capacitors and filters

Conclusions

Recall: Circuit elements

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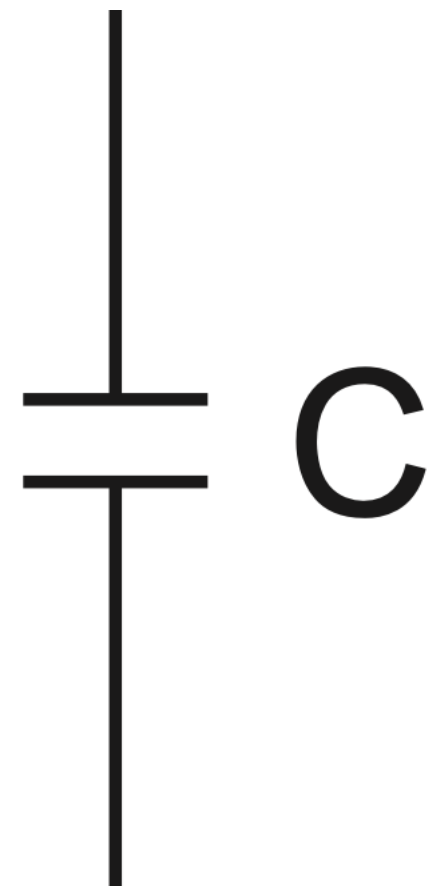
. How do they work? Passive vs active elements?



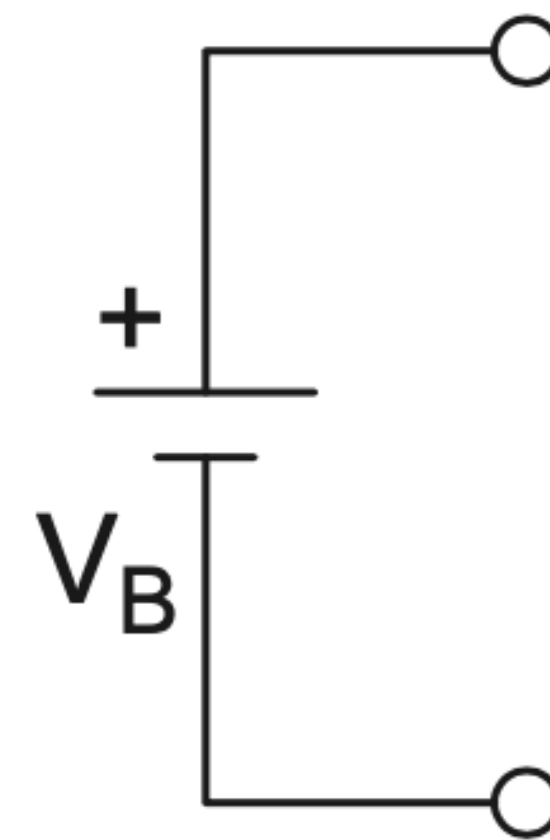
Resistor



Inductor

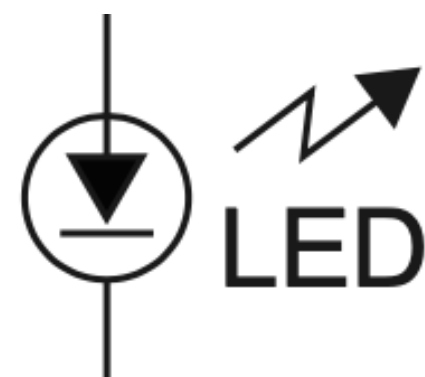


Capacitor



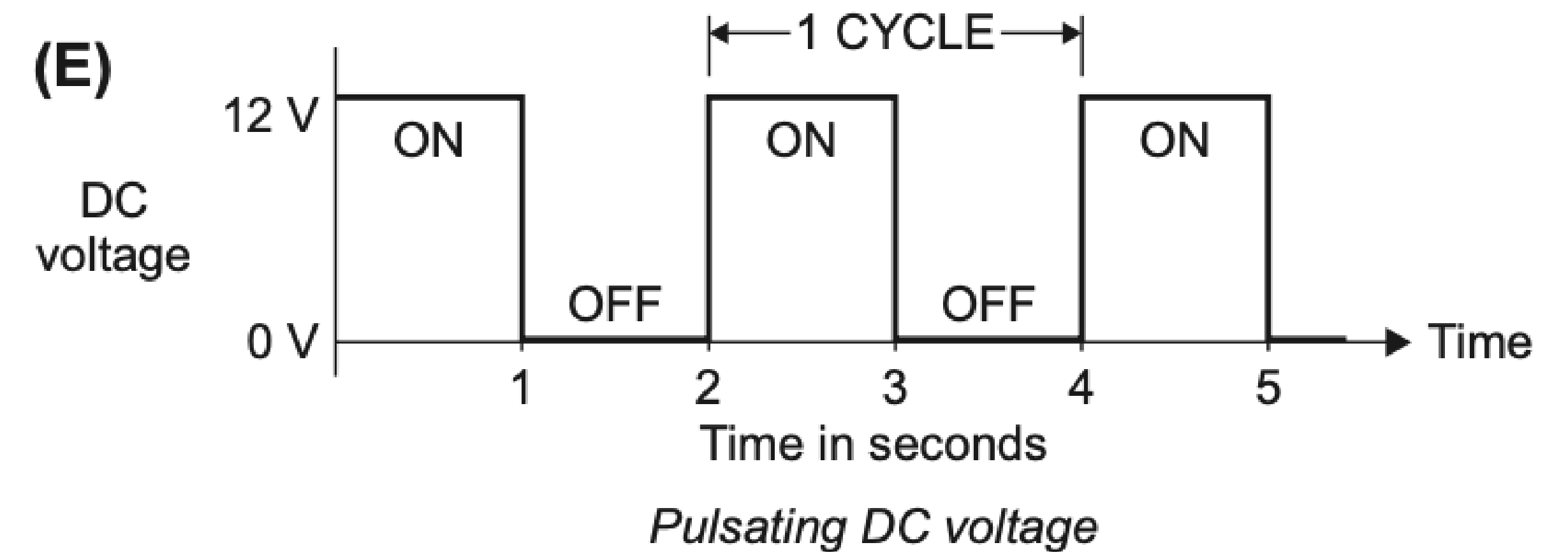
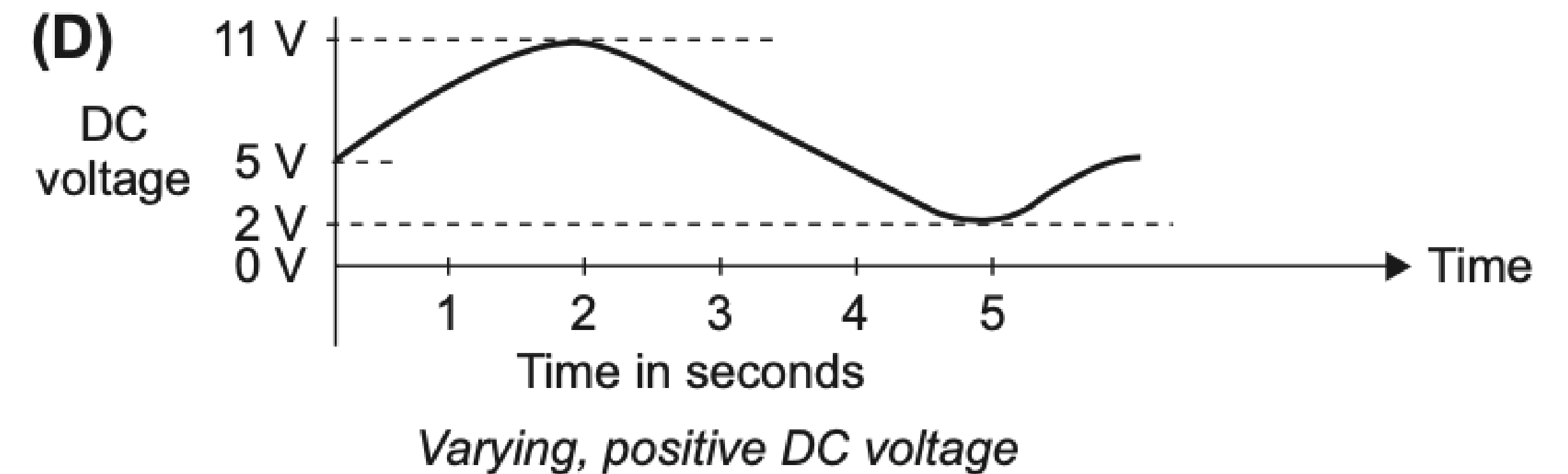
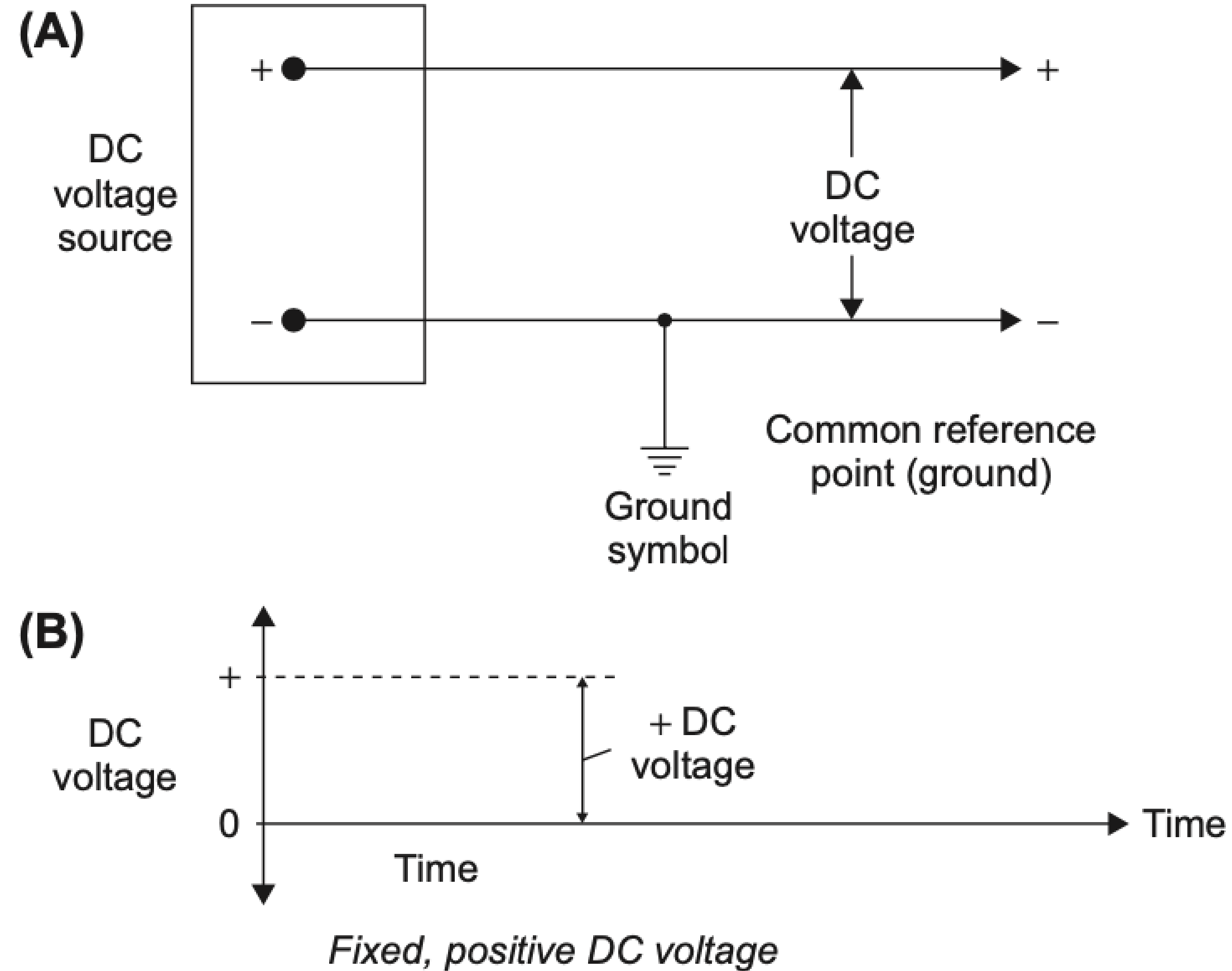
Voltage
Source

. What about a LED?



Recall: DC Voltage

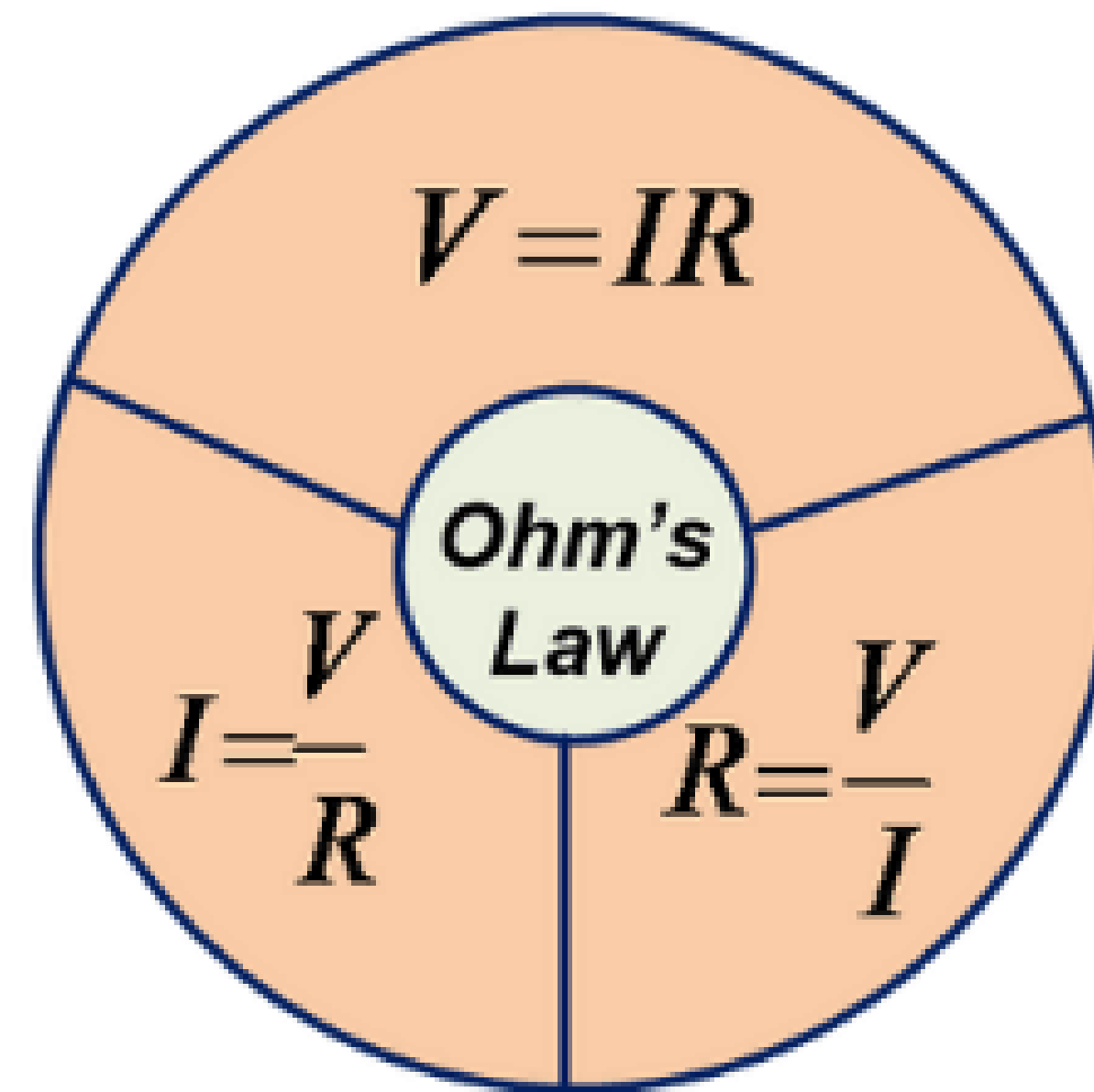
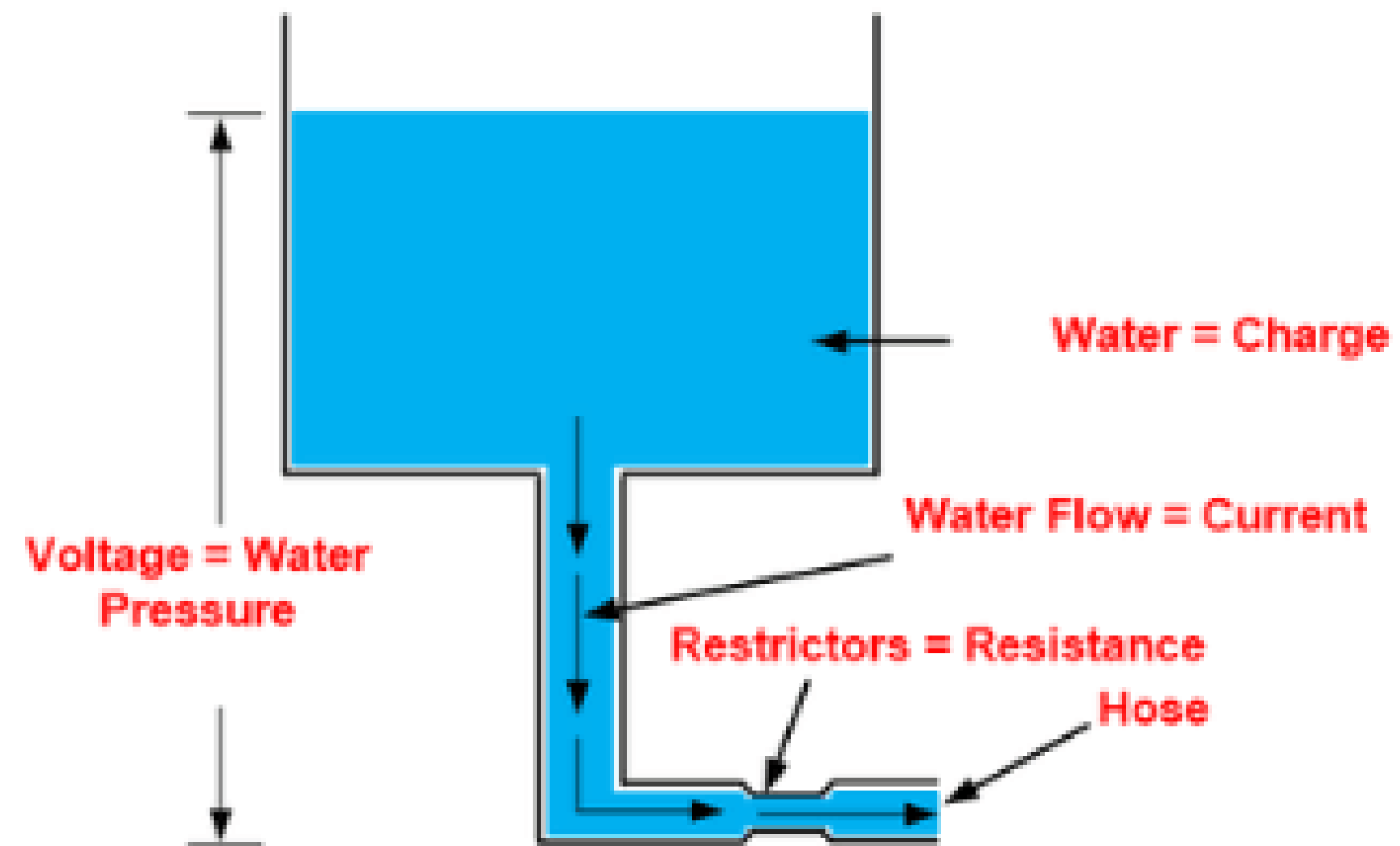
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Ohm's law

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What is Ohm's Law?

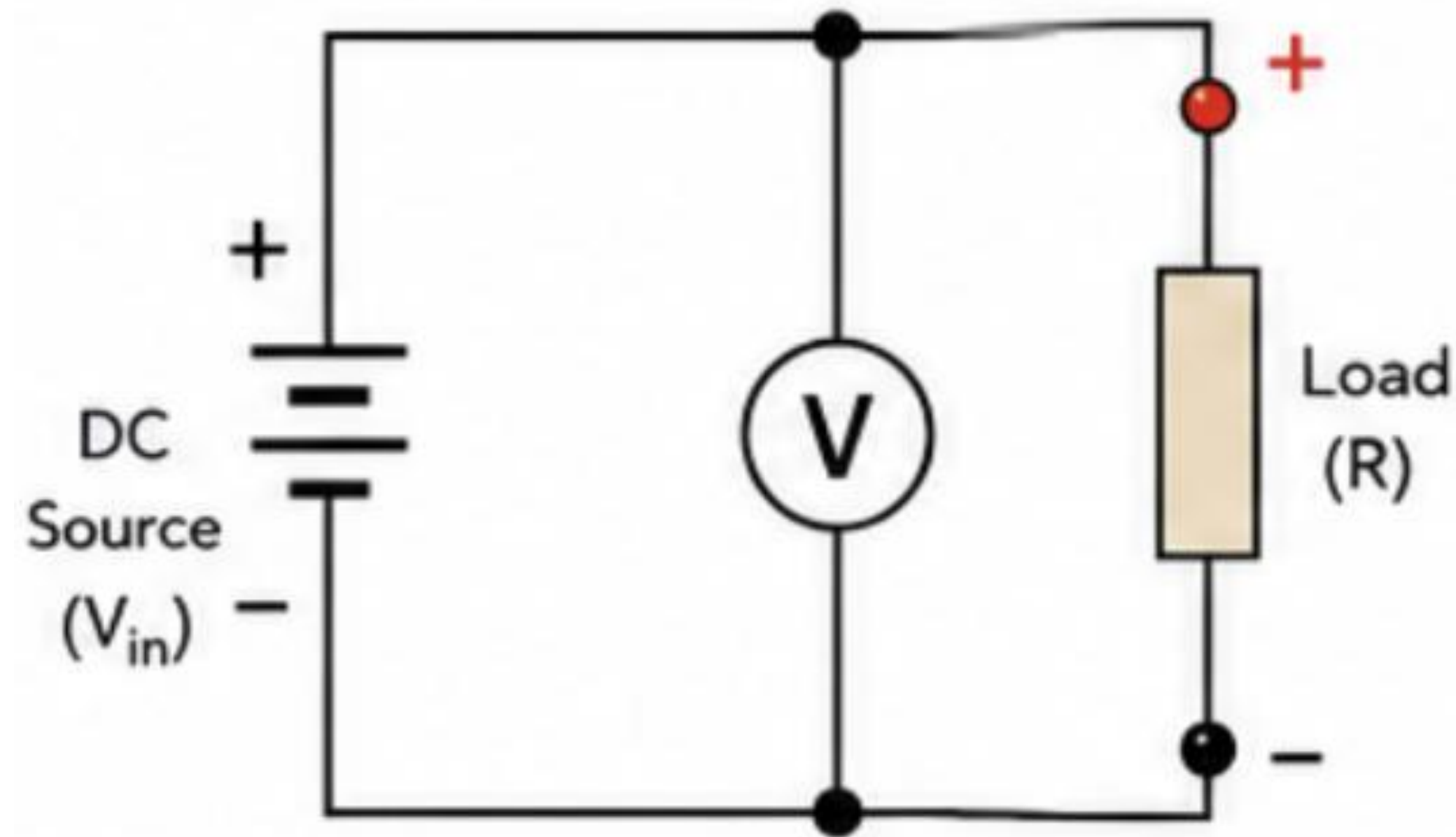


[ELECTRICAL4U.COM]

Voltage != Current

How an analog voltmeter works

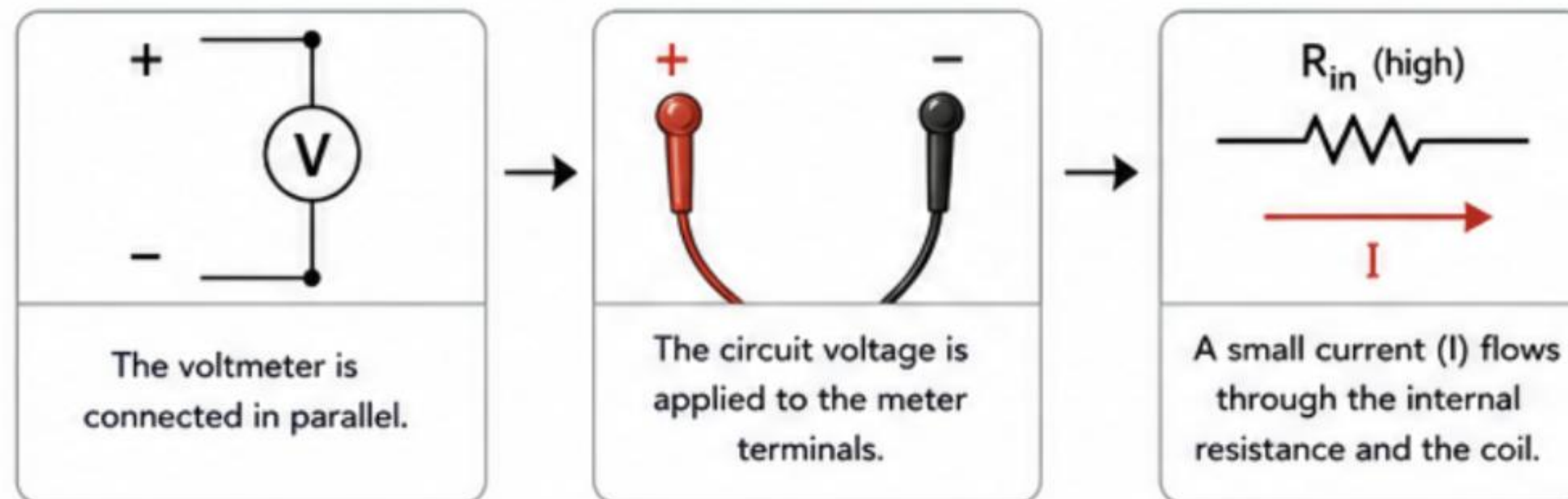
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The voltmeter is connected in parallel with the element whose voltage is being measure

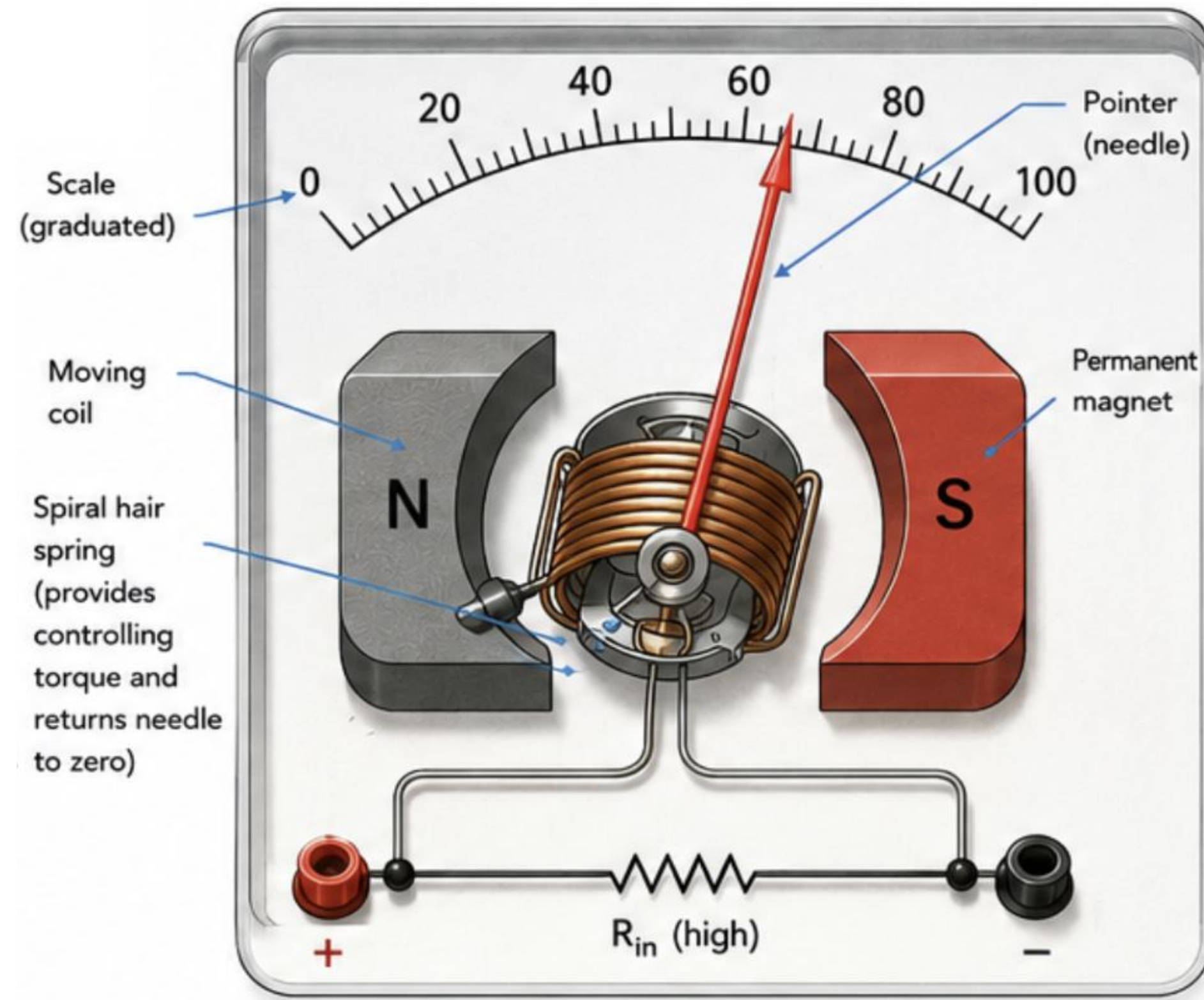
IMPORTANT:

The voltmeter has a very high internal resistance (R_{in}), so the current it draws is very small and does not disturb the circuit operation.



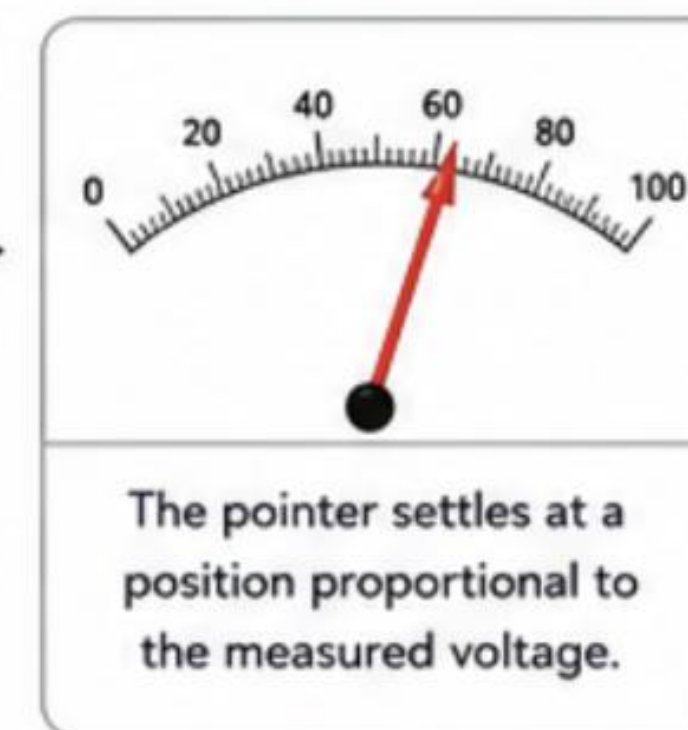
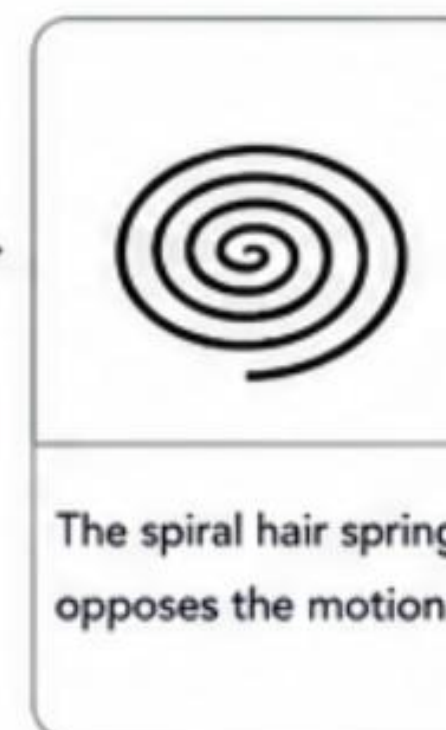
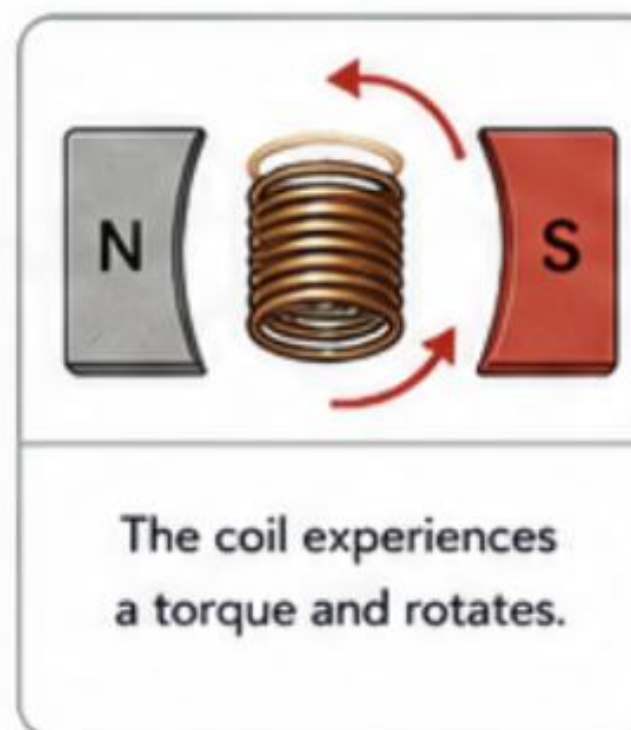
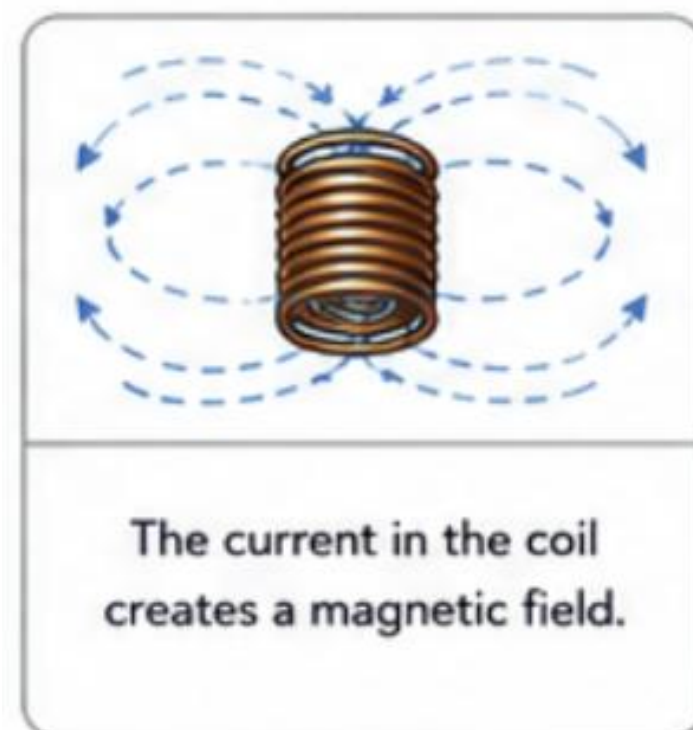
How an analog voltmeter works

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WHY DOES THE NEEDLE MOVE?

1. When a voltage is applied between the terminals, a small current flows through the internal resistance (R_{in}) and the moving coil.
2. The current in the coil produces a magnetic field.
3. This field interacts with the field of the permanent magnet, generating a torque (turning force).
4. The coil rotates, moving the pointer over the scale.
5. The spiral hair spring opposes the motion. The needle stops when the magnetic torque and the spring torque balance.



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Energy	Variables (Specific Fluctuation)	Common Measurements
Chemical	Chemical activity and/or concentration	Blood ion, O ₂ , CO ₂ , pH, hormonal concentrations and other chemistry.
Mechanical	Position Force, torque or pressure	Muscle movement, Cardiovascular pressures, muscle contractility Valve and other cardiac sounds
Electrical	Voltage (potential energy of charge carriers) Current (charge carrier flow)	EEG, ECG, EMG, EOG, ERG, EGG, GSR
Thermal	Temperature	Body temperature, thermography

- **Transducer is a devices that** converts energy from one form to another. In measurement systems, all transducers are so-called *input transducers*, they convert non-electrical energy into an electronic signal.
- In signal processing applications, **the purpose of energy conversion is to transfer information, not to transform energy.**
- In physiological measurement systems, all transducers are input transducers: **they convert non-electrical energy into an electronic signal.**

Example: Biotransducers

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The biotransducer is often the most critical element in the system: it often defines the accuracy or resolution of the measurement and acts as an interface between the life process and the rest of the system.

. This is not an stethoscope:

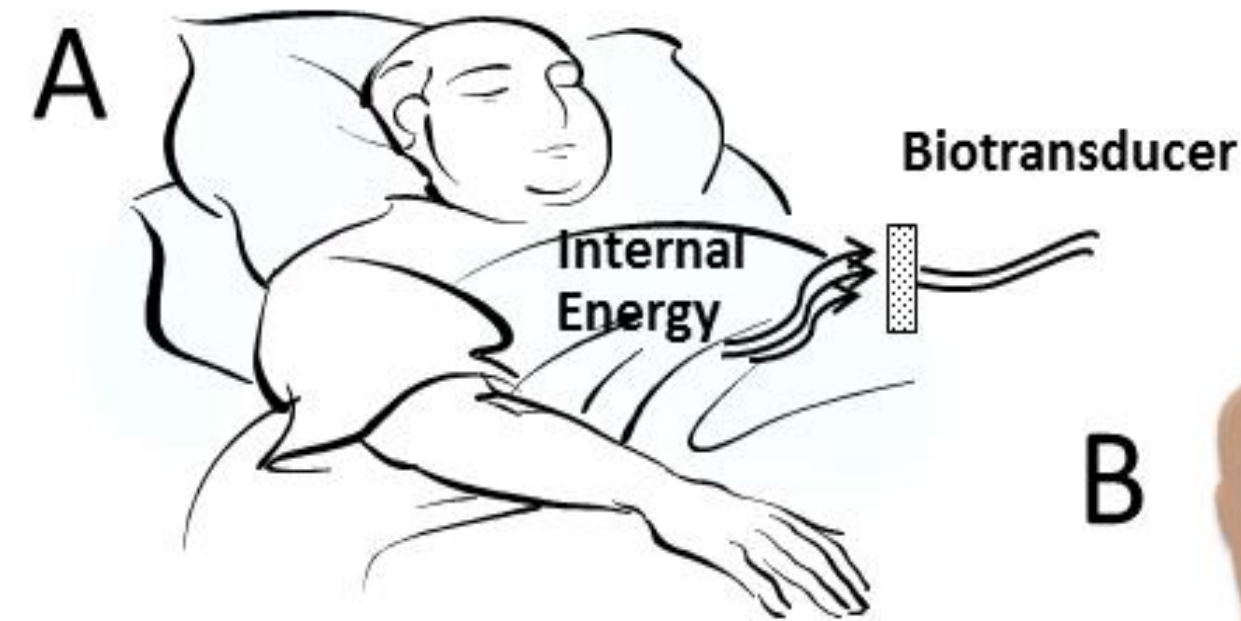


A cardiac microphone is a biotransducer that converts the mechanical sound energy produced by the heart into an electrical signal is shown. The device uses a piezoelectric disk that produces a voltage when it is deformed by movement of the patient's chest.

The white foam pad covers the piezoelectric disk and is specially designed to improve the coupling of energy between the chest and the piezoelectric disk.

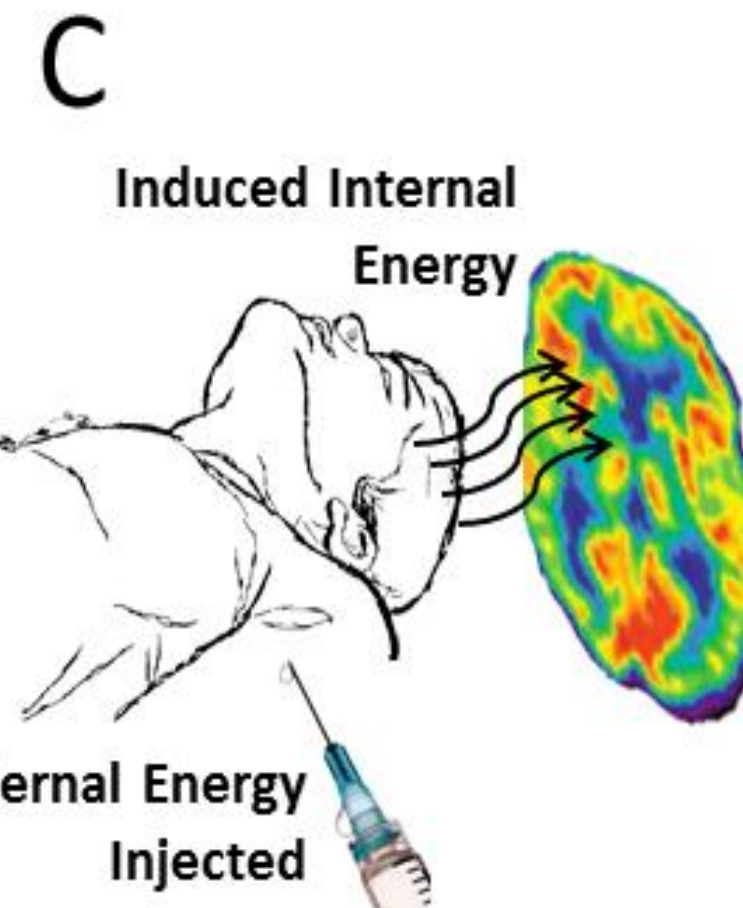
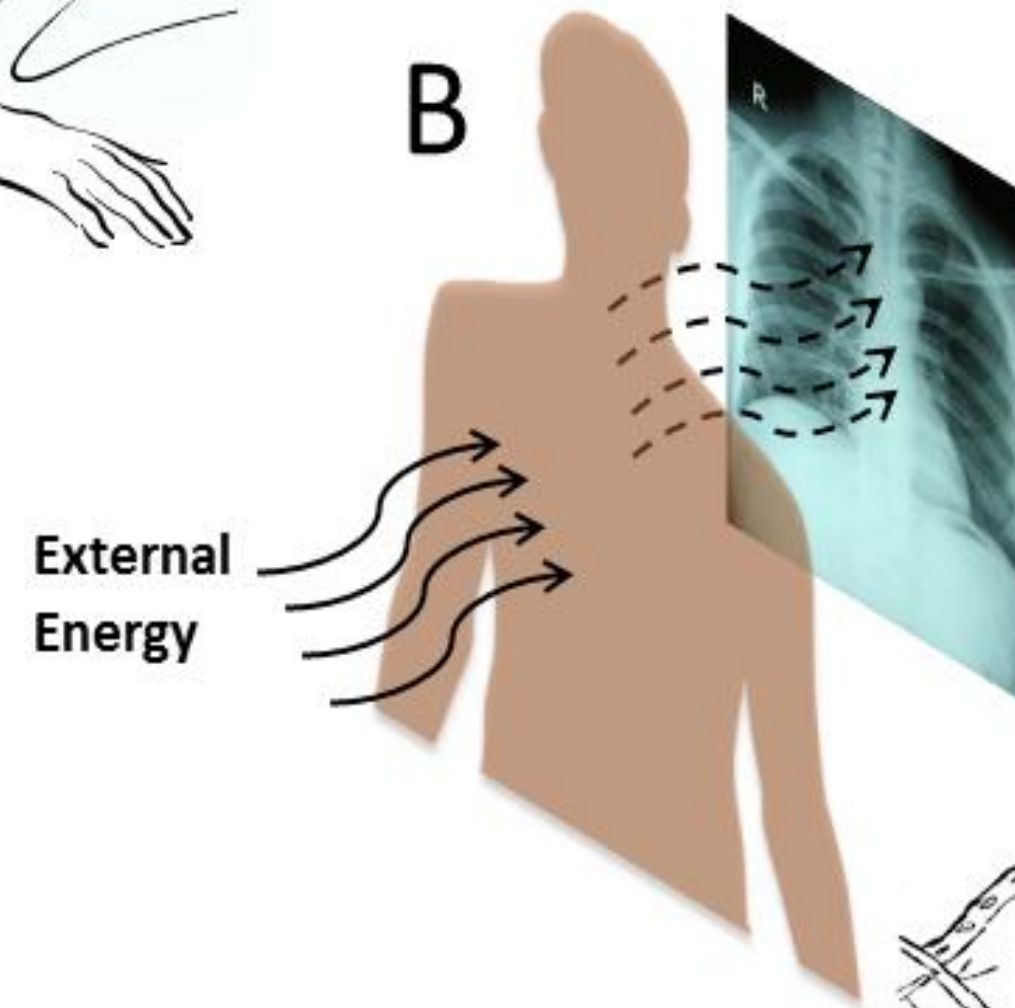
Three strategies for obtaining measurements from the body (or other physiological systems).

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A) The body generates the energy such as in heart sounds or electrically excitable tissue.

B) External energy is passed through the body which is differentially blocked (e.g., x-rays)



C) The body is converted to a signal source by an external substance or energy source (e.g. positron emission tomography..

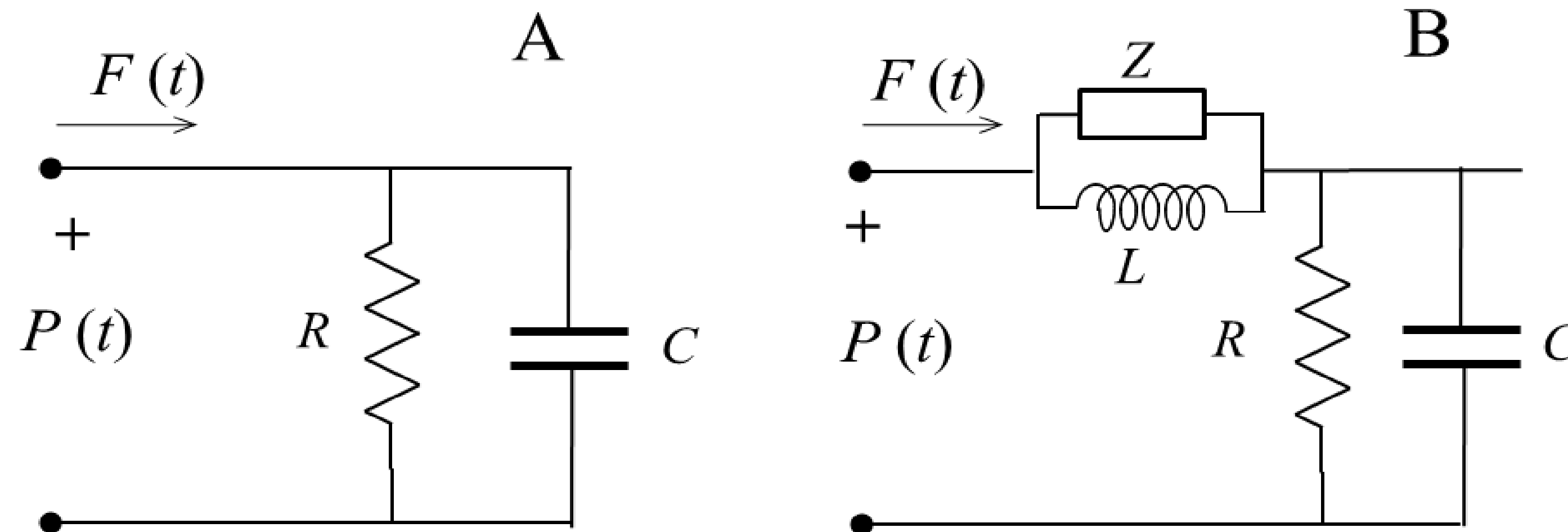
Example: Cardiac Analog Model

The original model, A, represented the **load on the heart as two passive elements**: a parallel resistor and a capacitor driven by pressure $P(t)$, the pressure generated by the heart.

The resistor, R , represents the **combined resistance of all the blood vessels (in units of mmHg sec/cm³)** although the primary contribution comes from the smaller peripheral vessels.

The capacitor, C , represents the **net compliance of the blood vessels (in units of cm³/mmHg)** which is mainly due to the aorta.

This is why understanding Ohm's law is important



The two-element Windkessel model was successful at modeling flow during diastole, but not systole, motivating the addition of other elements, B.

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Signal Encoding ~ Signal Representation

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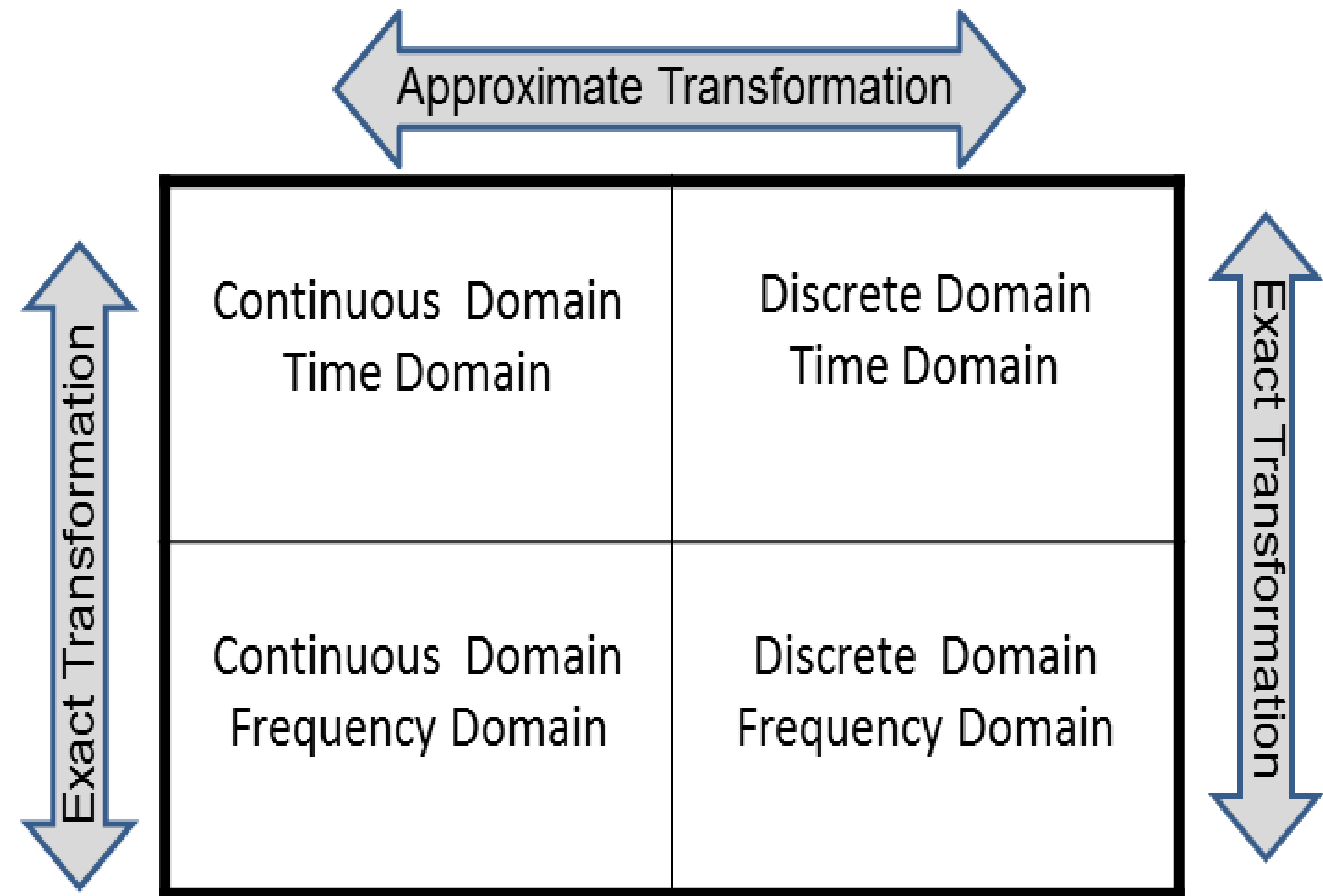
- Irrespective of the energy form or specific variable used to carry information, **some type of encoding scheme is necessary.**
- **Signals fall into two broad categories:** 1) one-dimensional signals where the dimension is usually time, and 2) two-dimensional signals where the dimensions are usually spatial.
- **Signals are represented in either of two domains:** 1) *continuous* and 2) *discrete*. These two domains are also termed *analog* and *digital*.

Signal Encoding

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A signal is represented in either the continuous or discrete domain, and in either the time or frequency domain.

- **Transformations between the time and frequency domain are exact:** there is no loss of signal information during the transfer.
- **Transformations between continuous and discrete domains are approximate** and only done in the time domain..



Continuous or Analog Domain Signals

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An analog signal mathematically encodes information as fluctuations in time. Such time-varying signals can be described by the equation:

$$x(t) = f(t)$$

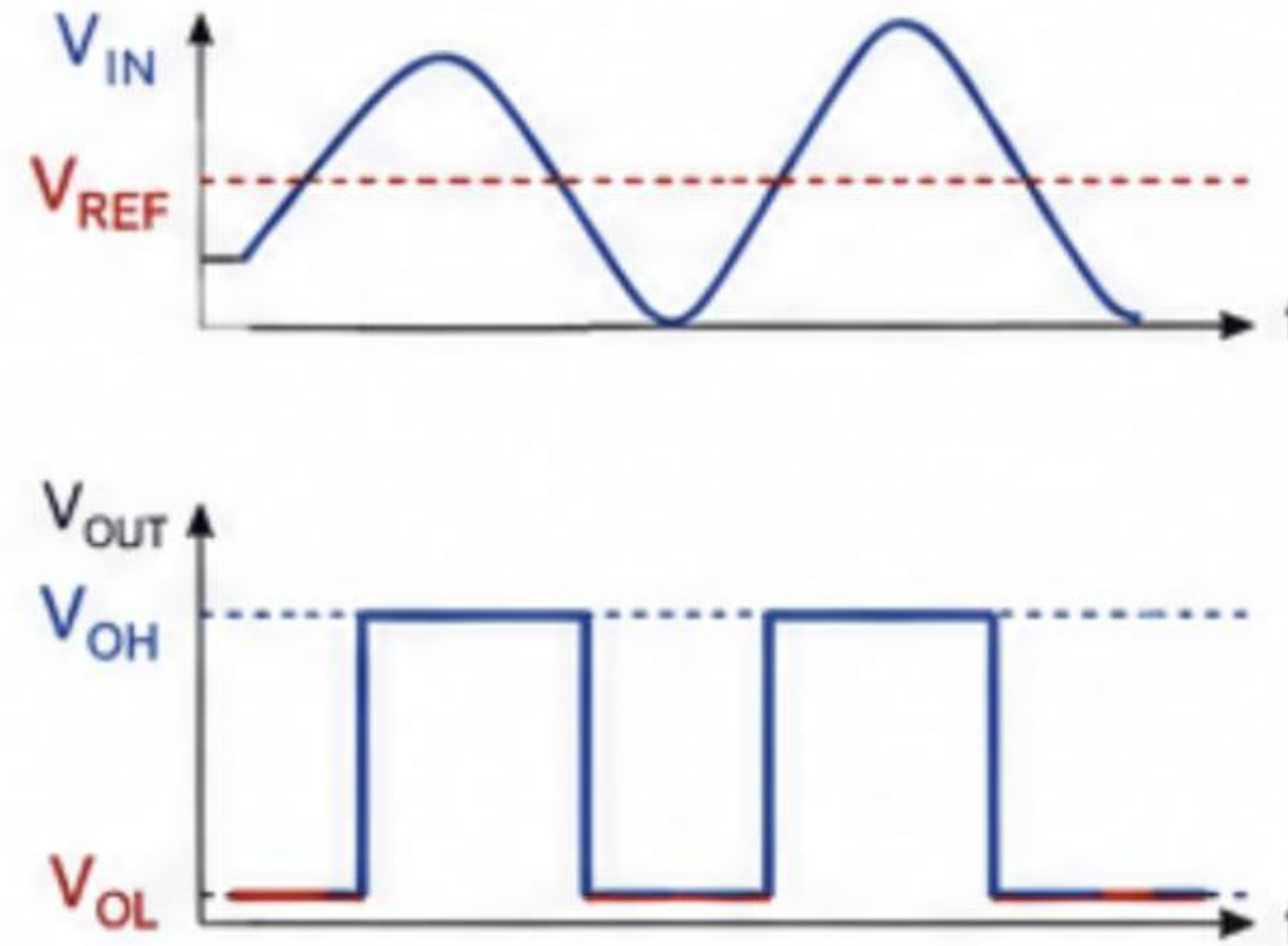
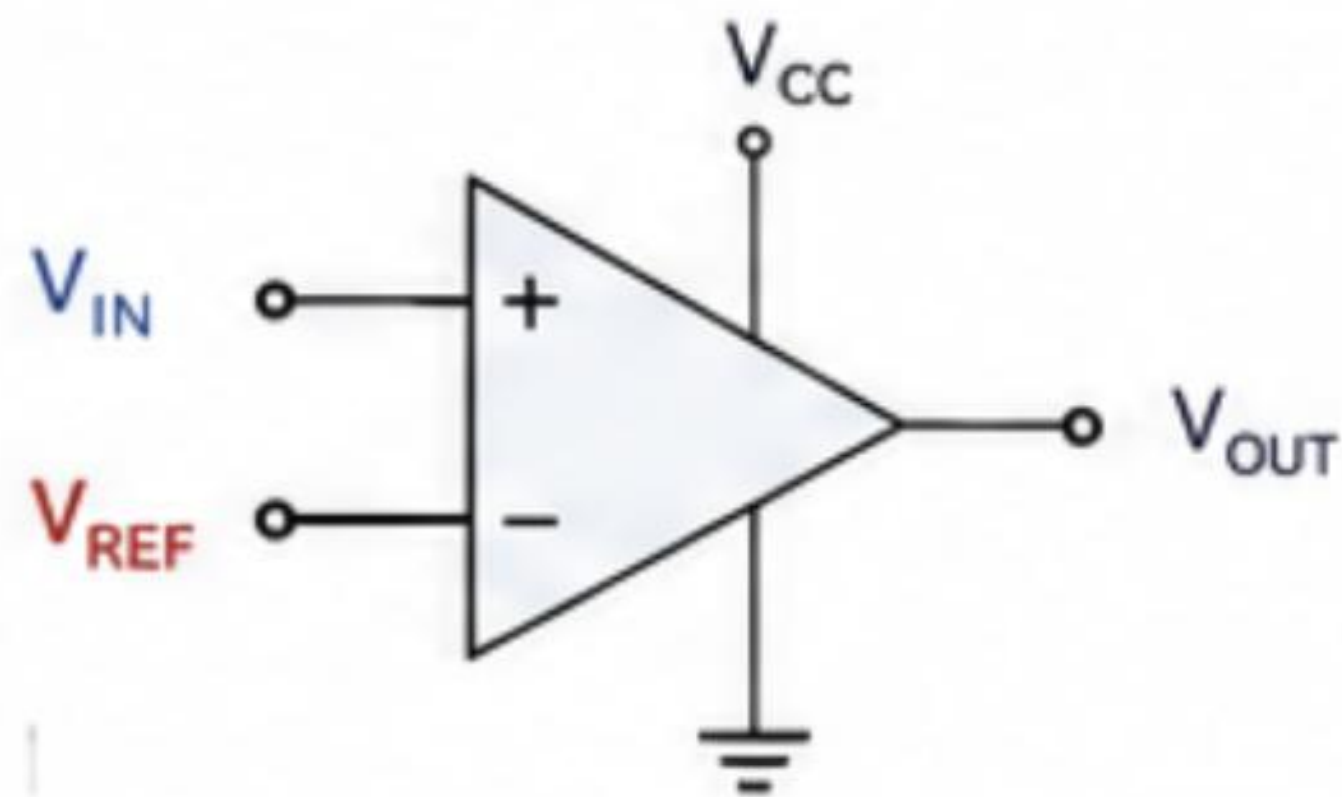
where $f(t)$ is some function of time and can be quite complex. For an electronic signal, this could be the value of the voltage or current at a given time.

Often an analog signal encodes the information it carries as a linear change in signal amplitude.

The comparator

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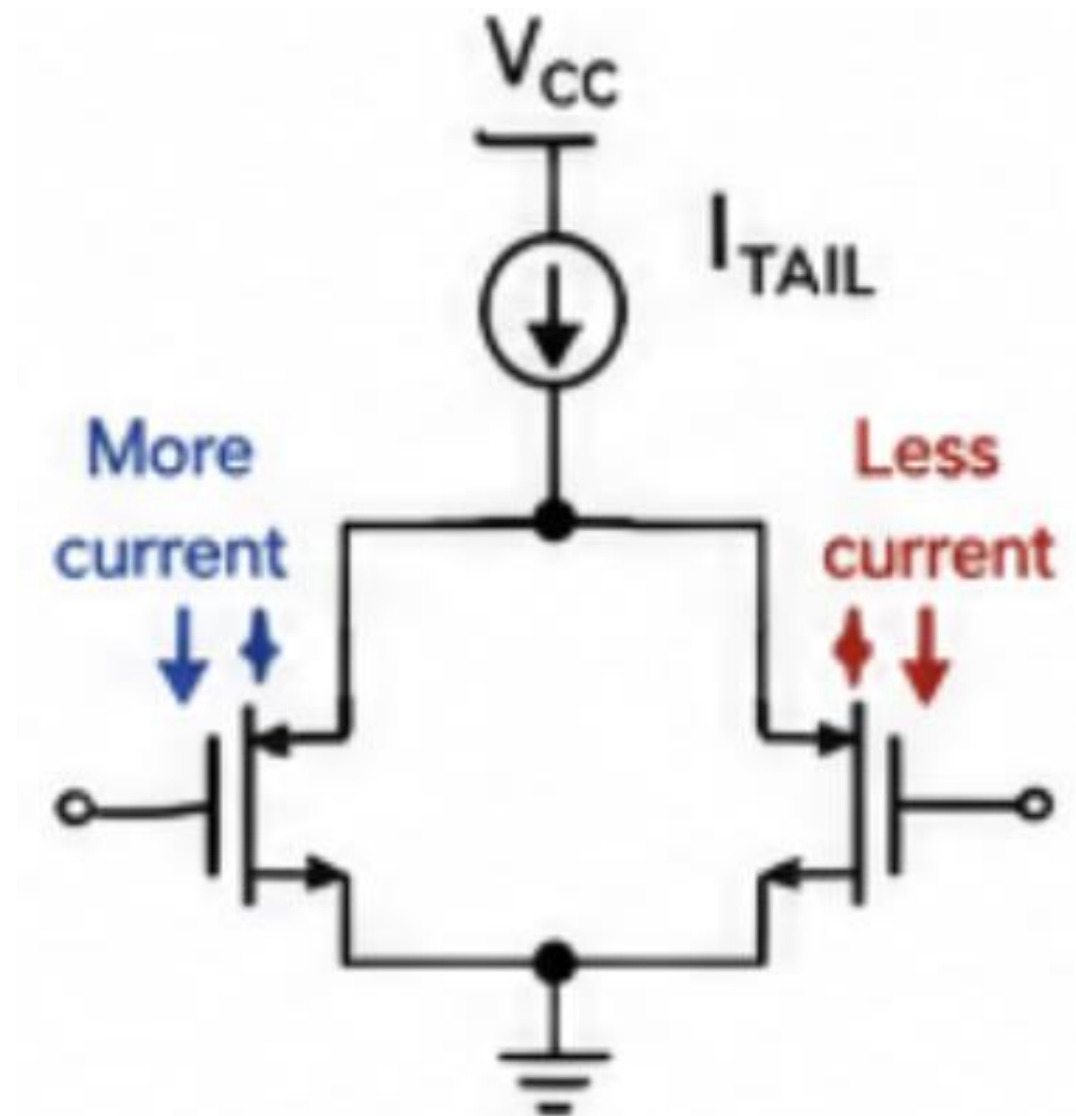
Comparator with a reference voltage.
It detects when V_{IN} crosses V_{REF} .



Operation:

- When $V_{IN} > V_{REF} \rightarrow V_{OUT} \approx V_{OH}$ (HIGH)
- When $V_{IN} < V_{REF} \rightarrow V_{OUT} \approx V_{OL}$ (LOW)

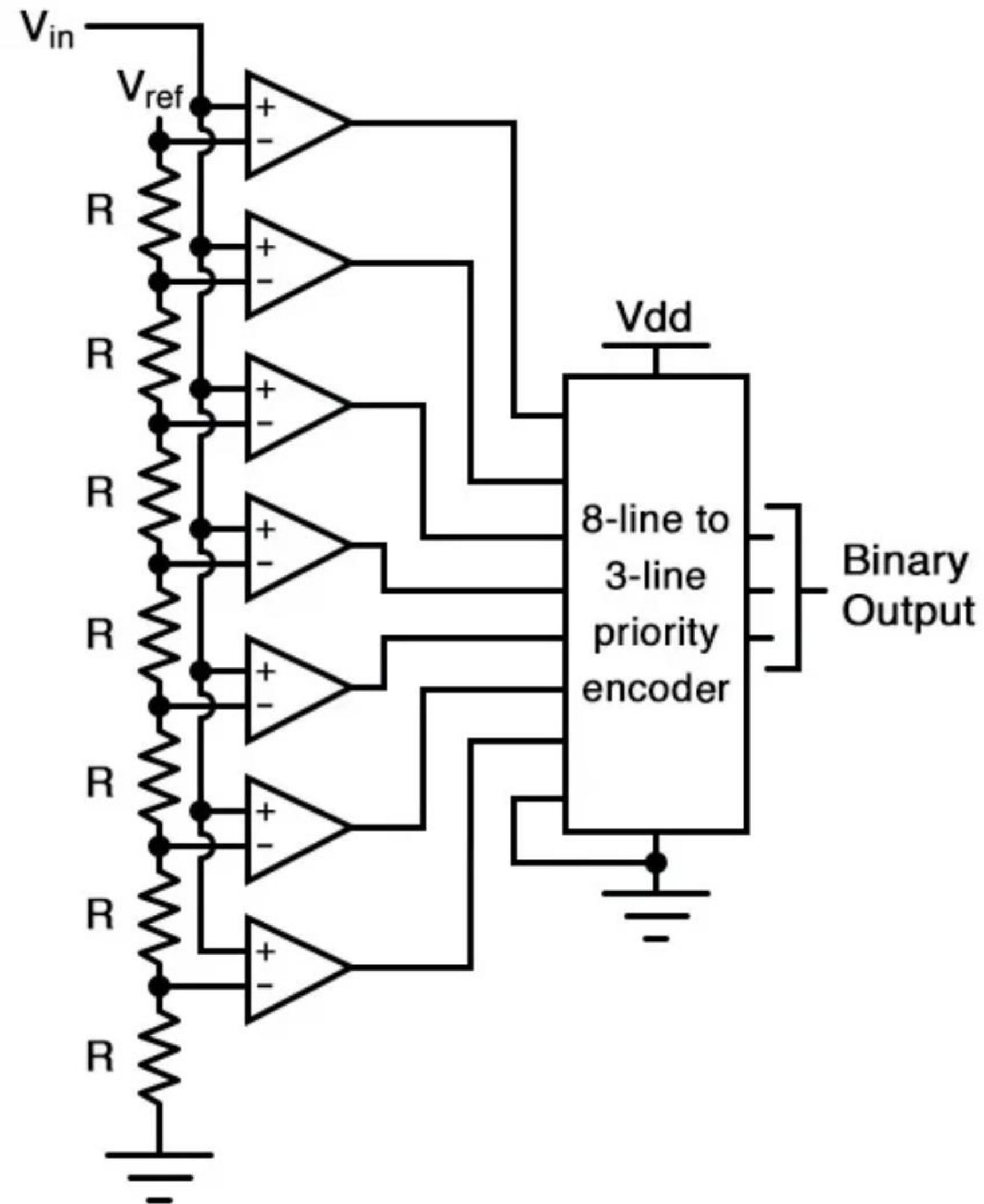
The output is a digital signal indicating the comparison result.



A tiny difference between V_+ and V_- creates a current imbalance in the input pair, which is amplified to saturate the output.

Flash ADC

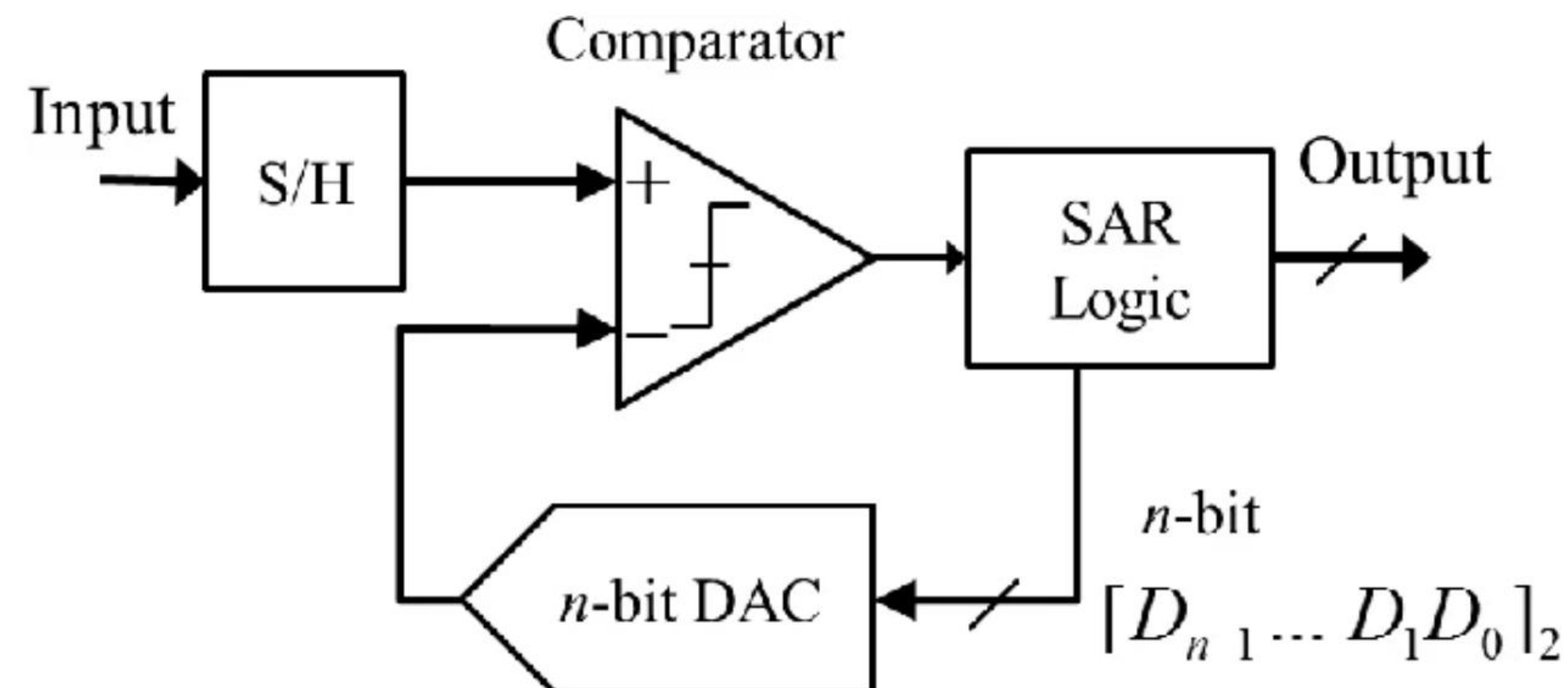
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Digital Output (D2 D1 D0)	Thermometer Code (C6 C5 C4 C3 C2 C1 C0)	Input Range (V_{IN})
000	0000001	0 to $1/8 V_{REF}$
001	0000011	$1/8$ to $2/8 V_{REF}$
010	0000111	$2/8$ to $3/8 V_{REF}$
011	0001111	$3/8$ to $4/8 V_{REF}$
100	0011111	$4/8$ to $5/8 V_{REF}$
101	0111111	$5/8$ to $6/8 V_{REF}$
110	1111111	$6/8$ to $7/8 V_{REF}$
111	1111111	$7/8$ to V_{REF}

Successive Approximation SAR ADC

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3. DETAILED EXAMPLE (3-BIT ADC)

Assume: $V_{IN} = 0.65 V_{REF}$

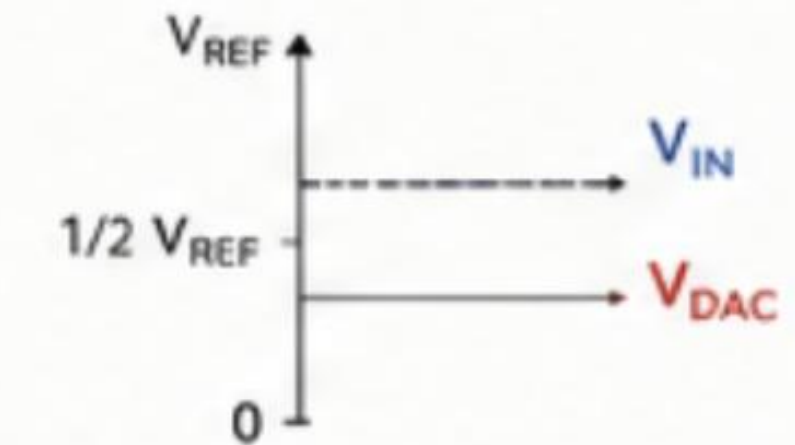
Step 1 (Test MSB)

Set D₂ = 1 → SAR = 100

DAC output = $1/2 V_{REF} = 0.50 V_{REF}$

$V_{IN} (0.65 V_{REF}) \geq 0.50 V_{REF} \rightarrow \text{Yes}$

Keep D₂ = 1



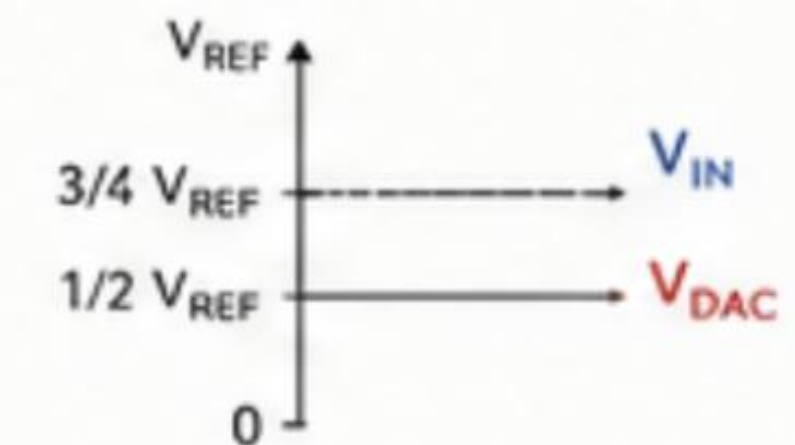
Step 2 (Test D1)

Set D₁ = 1 → SAR = 110

DAC output = $3/4 V_{REF} = 0.75 V_{REF}$

$V_{IN} (0.65 V_{REF}) < 0.75 V_{REF} \rightarrow \text{No}$

Set D₁ = 0 → SAR = 100



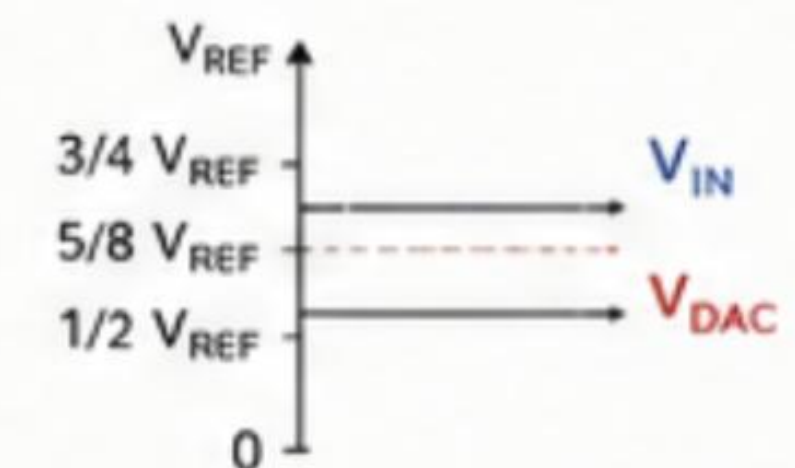
Step 3 (Test D0)

Set D₀ = 1 → SAR = 101

DAC output = $5/8 V_{REF} = 0.625 V_{REF}$

$V_{IN} (0.65 V_{REF}) \geq 0.625 V_{REF} \rightarrow \text{Yes}$

Keep D₀ = 1 → SAR = 101 (done)



Final Digital Output: D₂ D₁ D₀ = 1 0 1

Analog value $\approx 5/8 V_{REF} = 0.625 V_{REF}$ (close to $0.65 V_{REF}$)

Signal Encoding Example

A temperature transducer encodes room temperature into voltage as shown in the table.

Temperature (°C)	Voltage (volts)
-10	0.0
0.0	5.0
+10	10.0
+20	15.0

Then the encoding equation for temperature would be:

$$temperature = voltage * amplitude - 10$$

As defined later this is a linear encoding scheme.

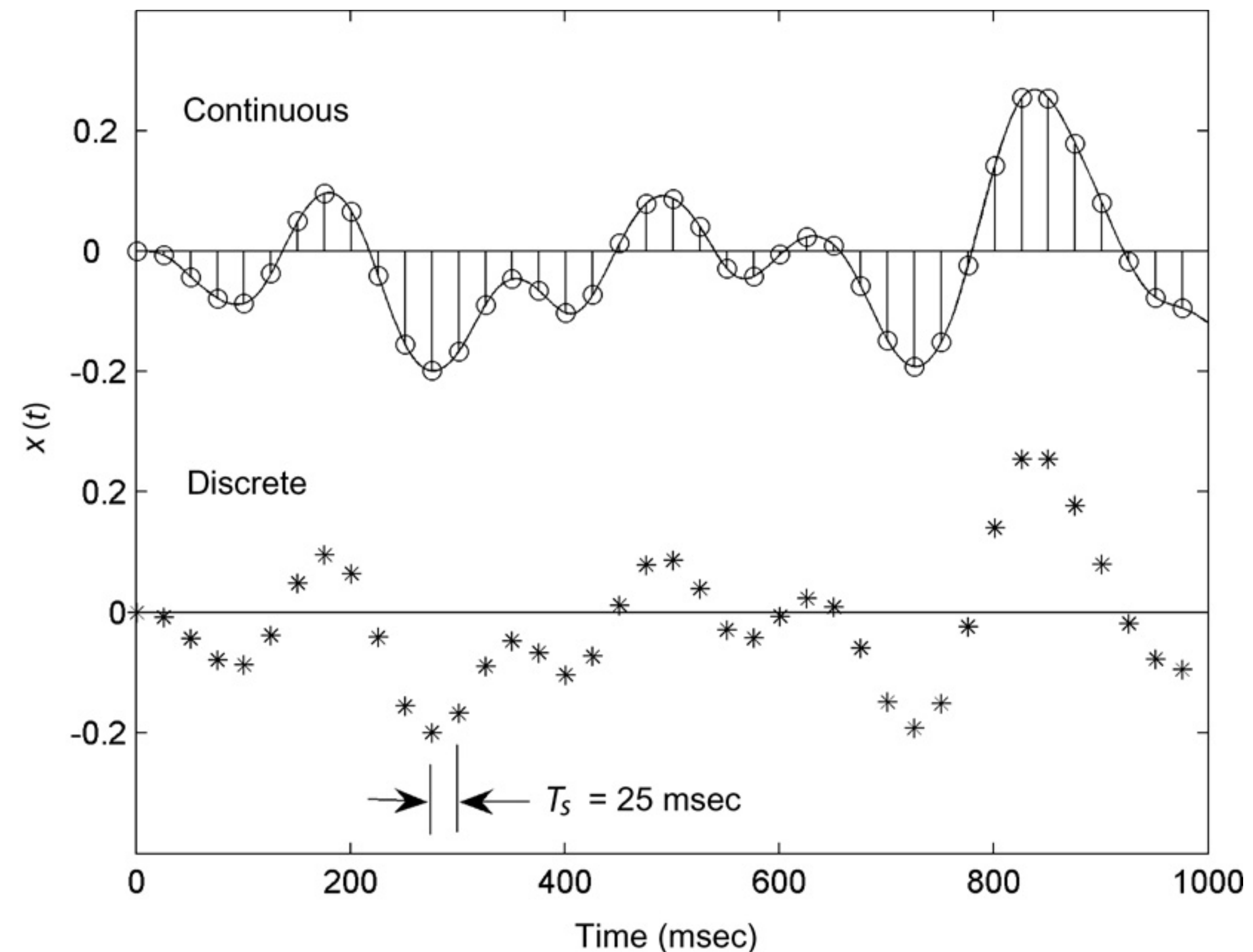
Discrete-Time or Digital Signals

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To be stored or processed in a digital computer, a continuous analog signal be converted to a series of numbers through a process **known as sampling**.

A continuous analog signal, $x(t)$, is sliced up into a sequence of digital numbers usually at equal time intervals, T_s .

T_s is known as the *sampling interval*.



Digital Signals

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- **These sequential numbers** represent the value of the analog signal at discrete points in time determined by the sample interval, T_s :

$$t = nT_s = n/f_s$$

Where f_s is the *sampling frequency*, $f_s = 1/T_s$ and where n is the position of the number in the sequence.

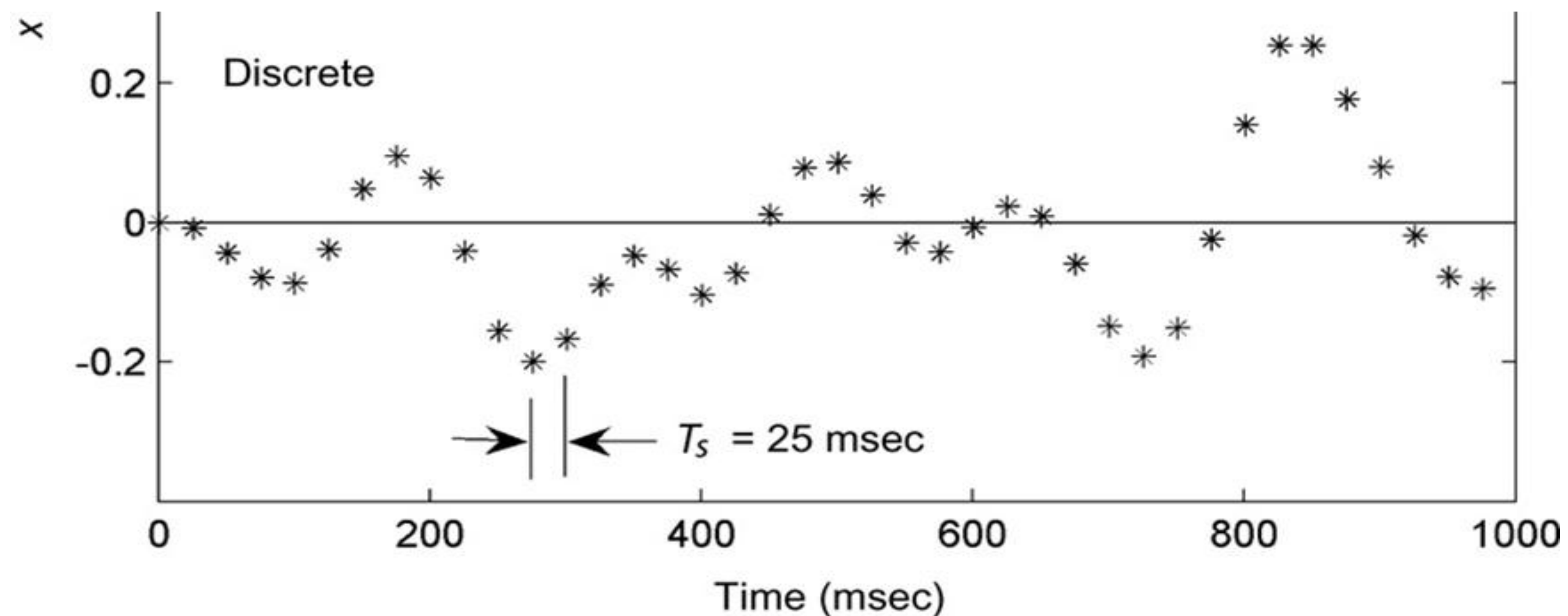
- **Usually this series of numbers would be stored in sequential memory locations (called an array or vector) with $x[1]$ followed by $x[2]$, then $x[3]$, and so on.**

Sampled or Digitized Signals

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- **A sampled time signal**, also referred to as a digitized signal or simply digital signal, can be easily stored in a digital computer.
- A digital signal, $x[k]$, is just a series of numbers:

$$x[k] = x_1, x_2, x_3, \dots, x_N.$$



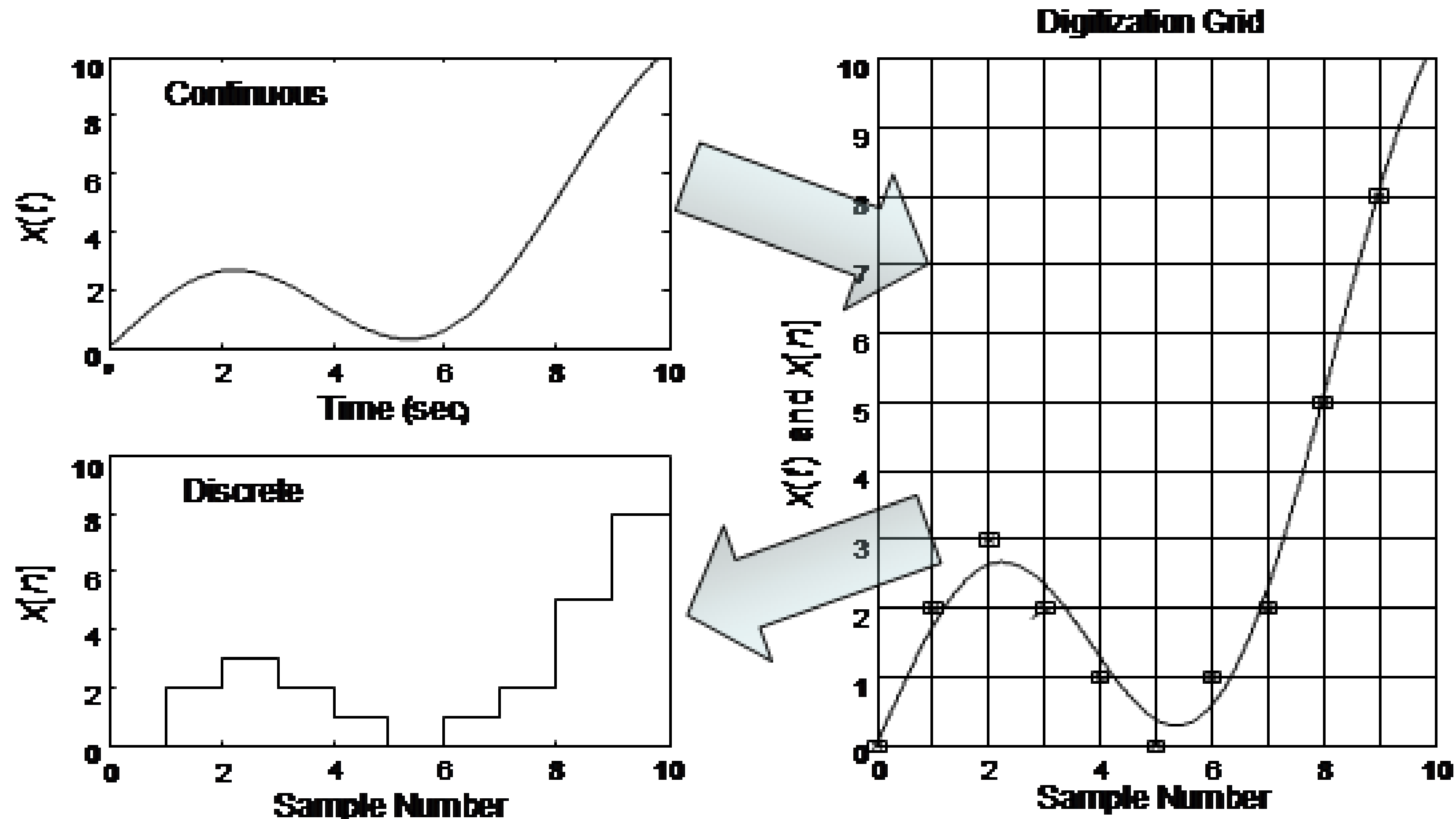
In this signal:

$x[n] = 50.0, 0.81, 0.96, 0.41, 20.3, 20.6, 20.3, 0.18, 0.34, 20.0, 20.5, 20.6, 20.0, 0.71, 0.99, 0.52).$

Amplitude Slicing: Quantization

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- Since digital numbers can only represent discrete or specific amplitudes, the analog signal must also be sliced up in amplitude.



To *digitize* or *sample* an analog signal requires slicing the signal in two ways: **in time** and **in amplitude**.

Analog-to-digital Conversion

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Example 1 A 12-bit analog-to-digital converter (ADC) advertises an accuracy of $\pm$ the least significant bit (lsb). If the input range of the ADC is 0 to 10 volts, what is the accuracy of the ADC in analog volts?

Analog-to-digital Conversion

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Example 1 A 12-bit analog-to-digital converter (ADC) advertises an accuracy of $\pm$ the least significant bit (lsb). If the input range of the ADC is 0 to 10 volts, what is the accuracy of the ADC in analog volts?

Solution: If the input range is 10 volts, then the analog voltage represented by the lsb would be:

$$V_{LSB} = \frac{V_{\max}}{2^{\text{Nu bits}}} = \frac{10}{2^{12}} = \frac{10}{4096} = .0024 \text{ volts}$$

Hence the accuracy would be $\pm .0024$ volts.

Example 2:

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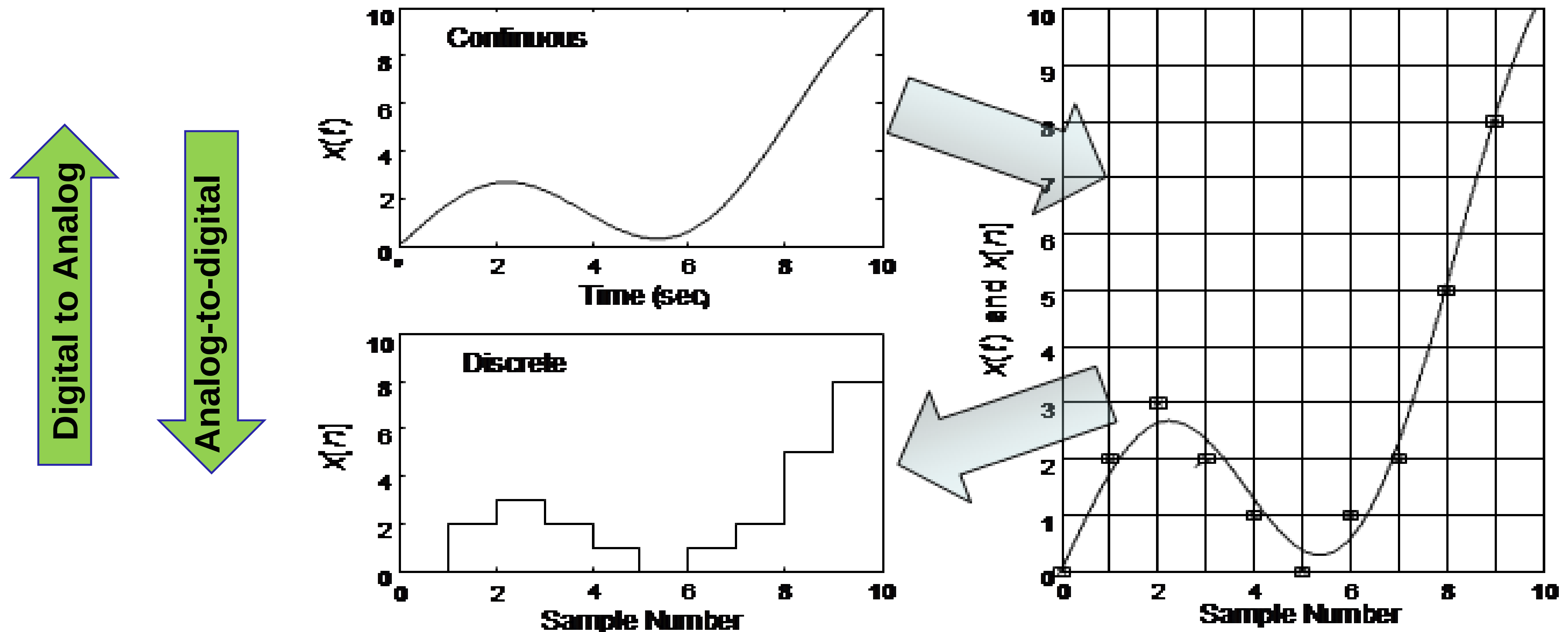
- . Given an analog output signal whose voltage should range from 0 to 10 V, and an 8-bit digital encoding, provide the encodings for the following desired voltages: (a) 0 V, (b) 1 V, (c) 5.33 V, (d) 10 V, (e) What is the resolution of our conversion?

Analog-to Digital Conversion

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Conversion between the analog and digital domain is done with special hardware devices known as analog-to-digital or digital-to-analog converters (ADC/DAC).

Continuous (analog) signal $\leftrightarrow$ Discrete (digital) signal
Digitization Grid



Is a signal that has been transformed from the continuous to the discrete domain the same?

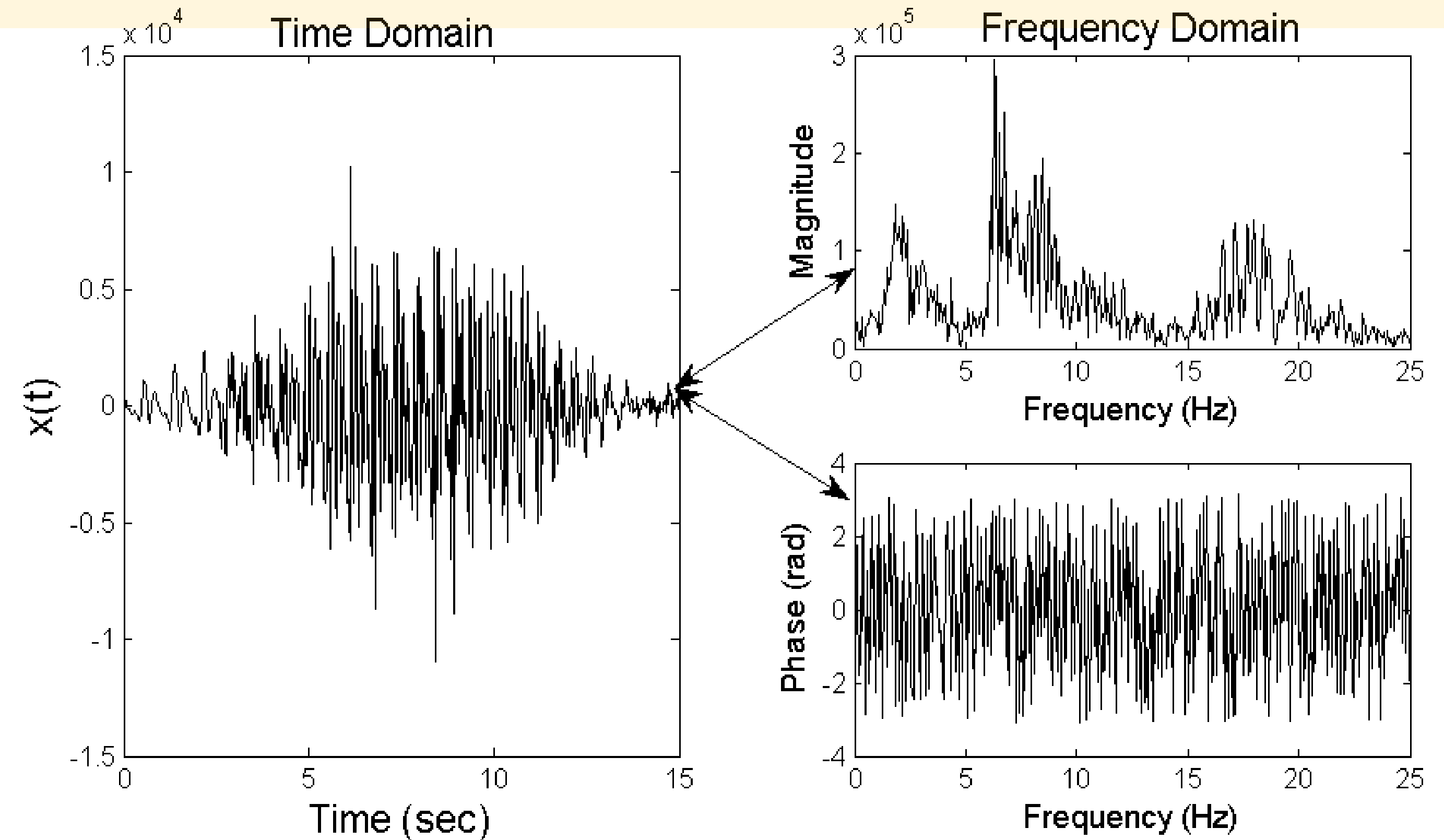
- Clearly not; as seen in the difference in the two different signals in the previous slide. Yet in signal analysis we often operate on discrete signals converted from an analog signal with the expectation (or assumption) that the discrete version is essentially the same as the original continuous signal.
- If they are not the same is there at least some meaningful relationship between the two? For now we will assume that all computer-based signals used in examples and problems are accurate representations of their associated continuous signals.

Time and frequency domain

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Signals are represented in the frequency domain as two functions of frequency: a magnitude and a phase.

Example: A section of EEG (electroencephalogram) signal in the time domain (left) and frequency domain (right).



- The time domain representation is not easy to interpret, nor is the frequency phase domain.
- The magnitude does show some interesting features: peaks around 1-3 Hz and 16-18 Hz, and a large peak around 7-8 Hz. These peaks in frequency are well known to neurologists.

Time and frequency domain

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- Because the frequency domain signals are functions of frequency, they are called *spectra*, the magnitude spectrum and phase spectrum.
- Although the time and frequency domain representation hold the same information they do not store it in the same way so that one or the other may be more diagnostically useful in a given situation.
- It is fairly easy to transform from one domain to the other using techniques such as FFT.

Transformada Discreta de Fourier

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$$X[k] = \sum_{n=0}^{N-1} x[n] \cdot e^{-j2\pi \frac{kn}{N}}$$

What does the DFT do?

- It **projects** the signal onto **complex waves** (complex exponentials) of different frequencies.
- It tells you **how much of each frequency** is present in the signal.
- The result $X[k]$ is a **complex number**, which you can interpret as:

$$X[k] = A_k \cdot e^{j\phi_k}$$

Transformada Discreta de Fourier

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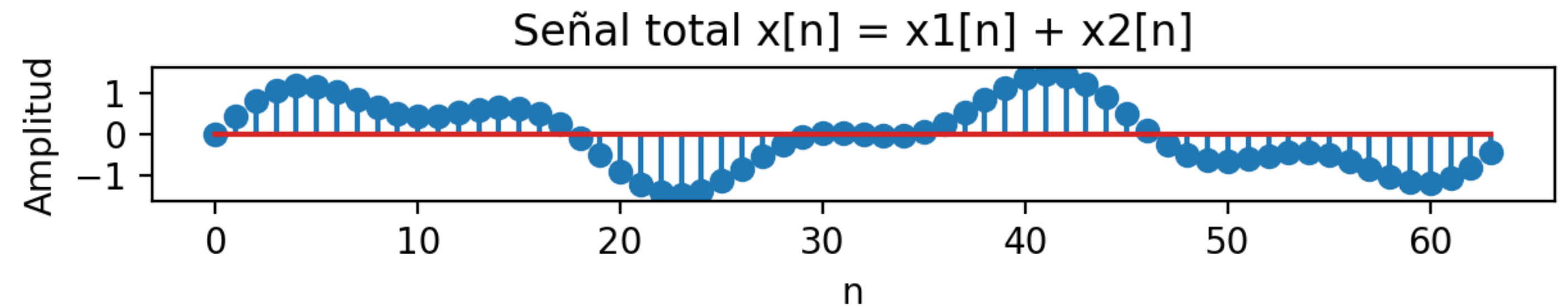
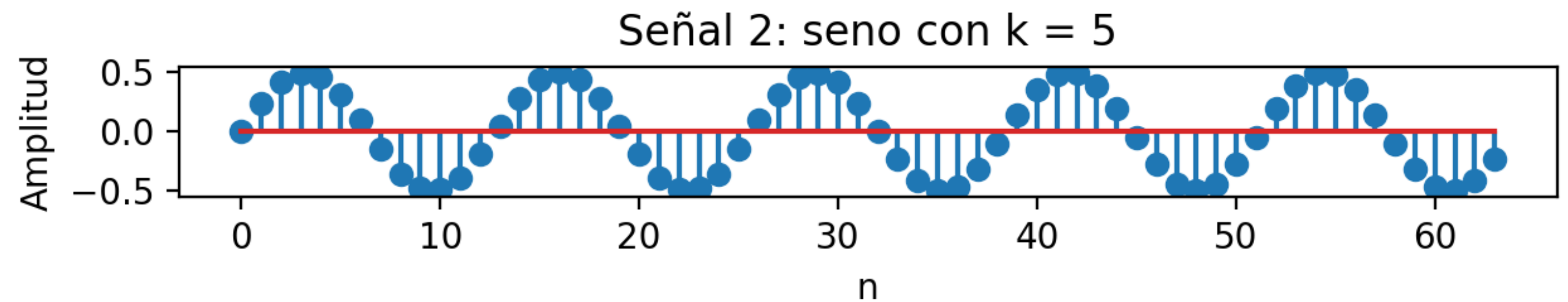
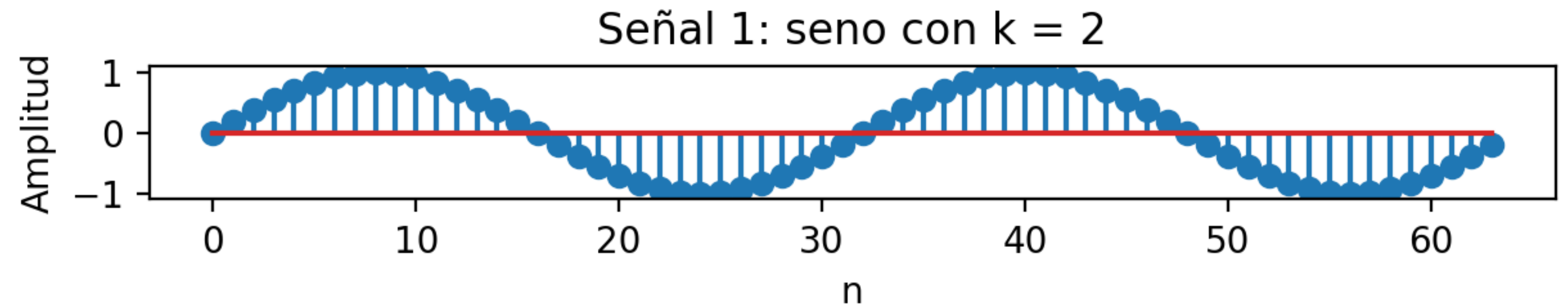
Señales individuales

```
x1=np.sin(2*np.pi*k1*n/N)
```

```
x2=0.5*np.sin(2*np.pi*k2*n/N)
```

Señal total

```
x = x1 + x2
```



Transformada Discreta de Fourier

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Señales individuales

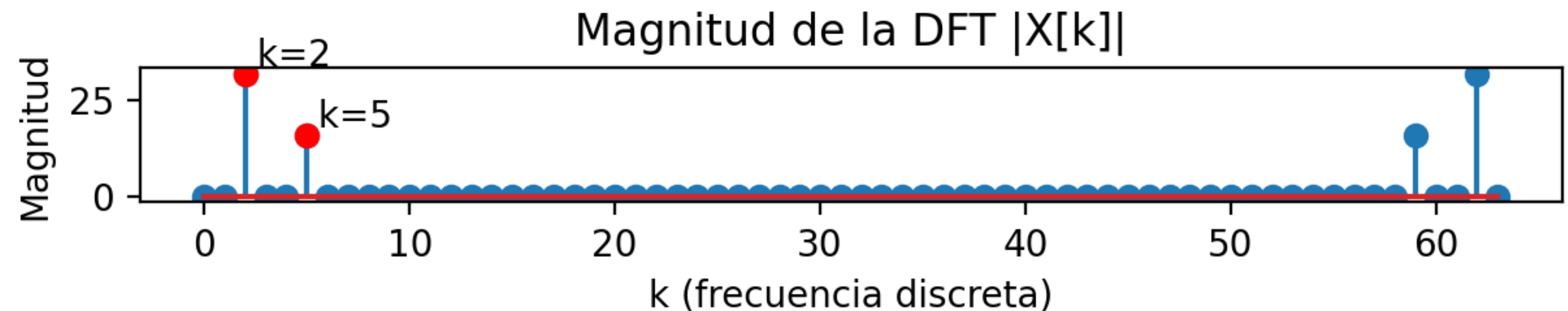
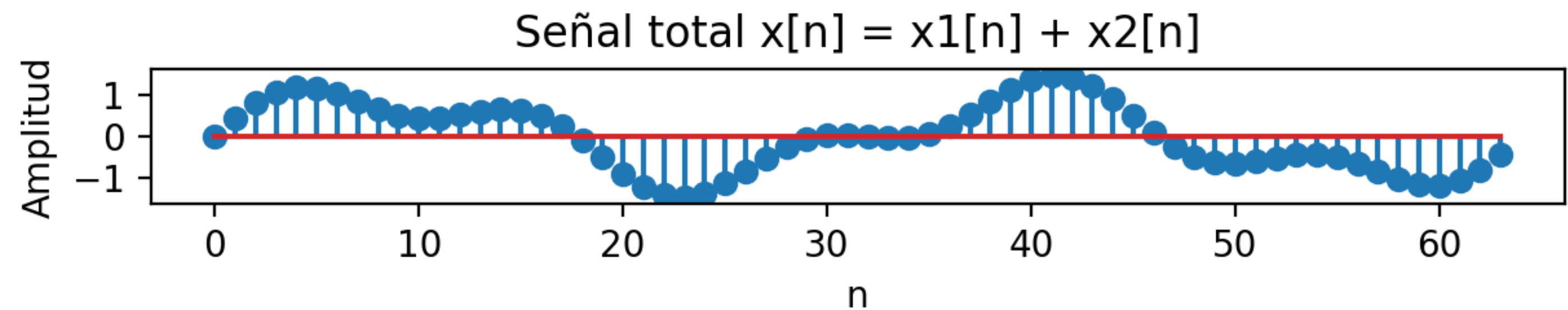
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Señal total

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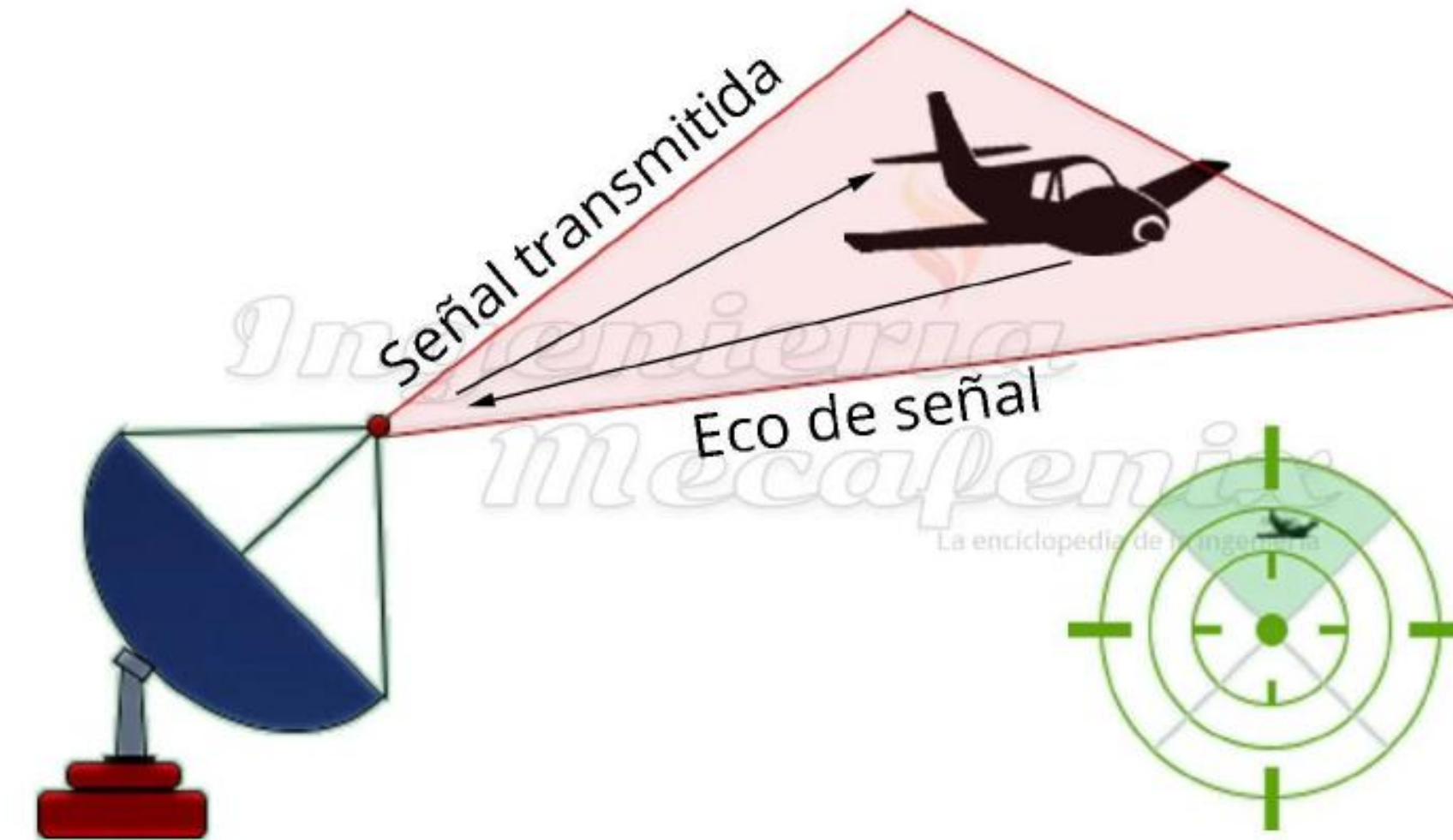
```
X = np.fft.fft(x)
```



Inverse DFT (frequency to time)

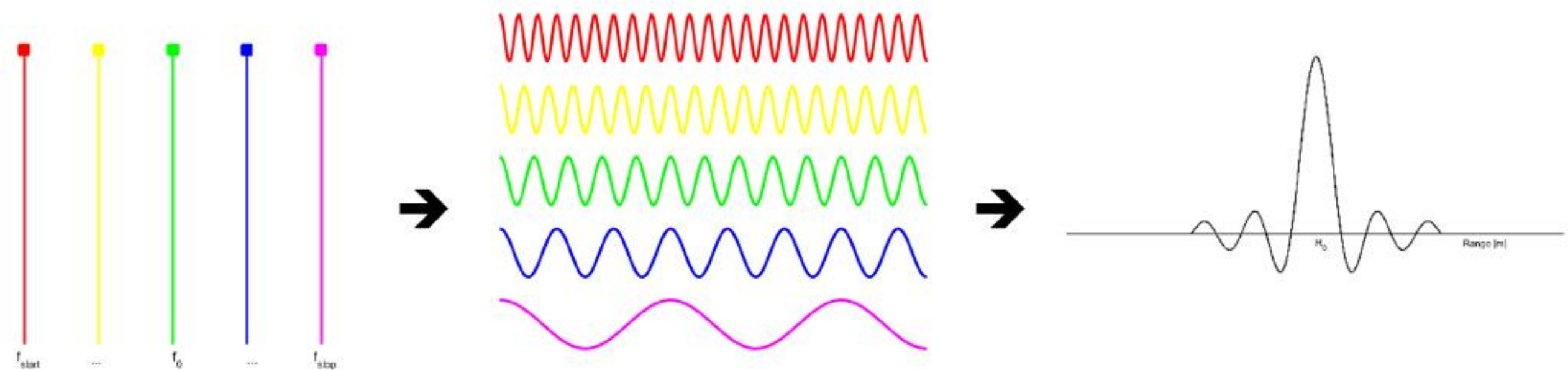
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In the same way we can apply the inverse transform, going from the frequency domain to the time domain.



Ingeniería Mecafenix

Application: SFCW radar



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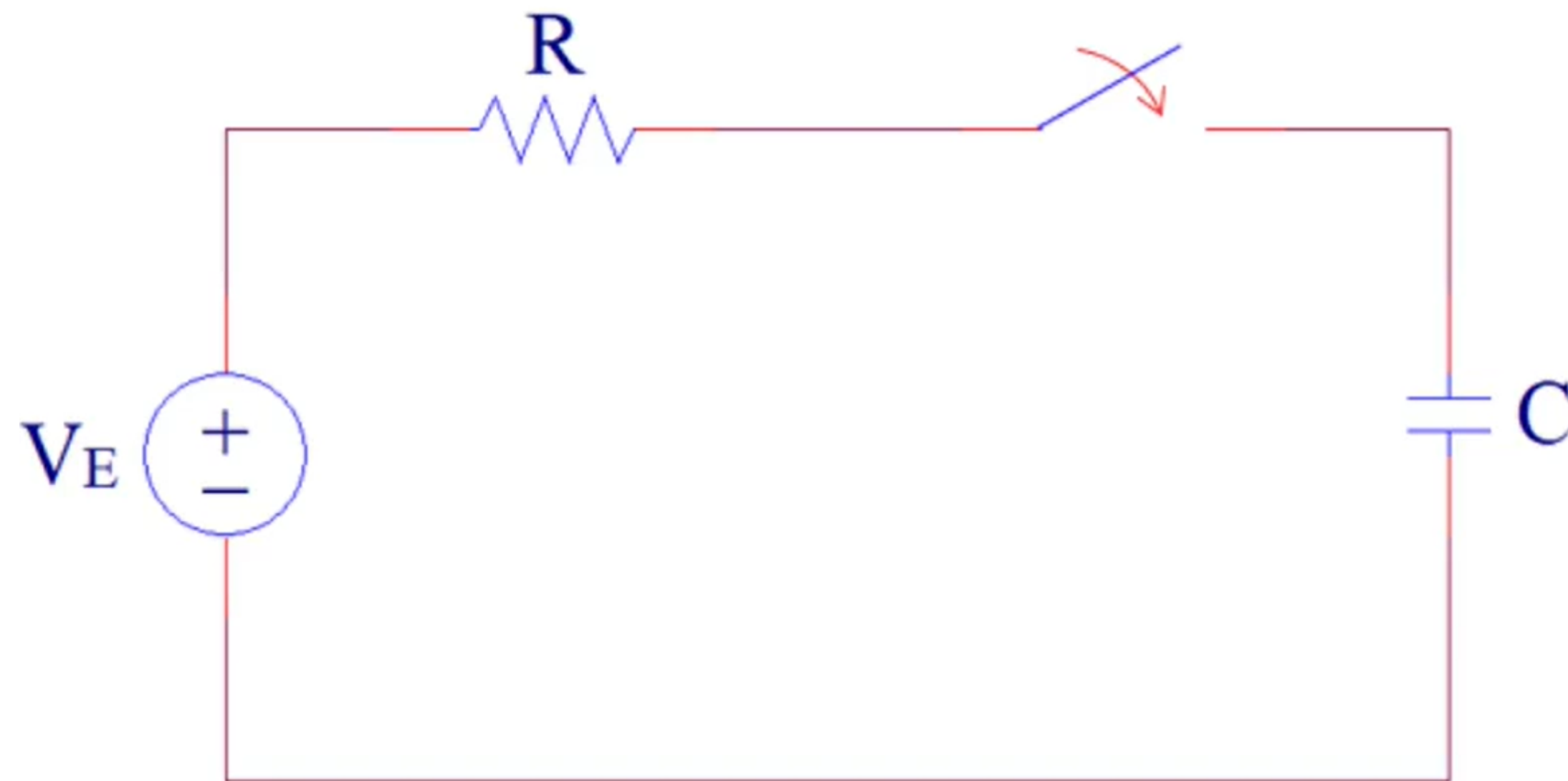
Carga de un capacitor

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Ecuación exponencial:

$$v(t) = V_E \left(1 - e^{-\frac{t}{\tau}}\right)$$

$$\tau = RC$$



Cargado

$$t = 0$$

$$v(0) = V_E (1 - e^0)$$

$$v(0) = V_E (1 - 1)$$

$$v(0) = 0$$

$$t = \tau$$

$$v(\tau) = V_E (1 - e^{-1})$$

$$v(\tau) = V_E (1 - 0,368)$$

$$v(\tau) = 0,632 V_E$$

$$t = 5\tau$$

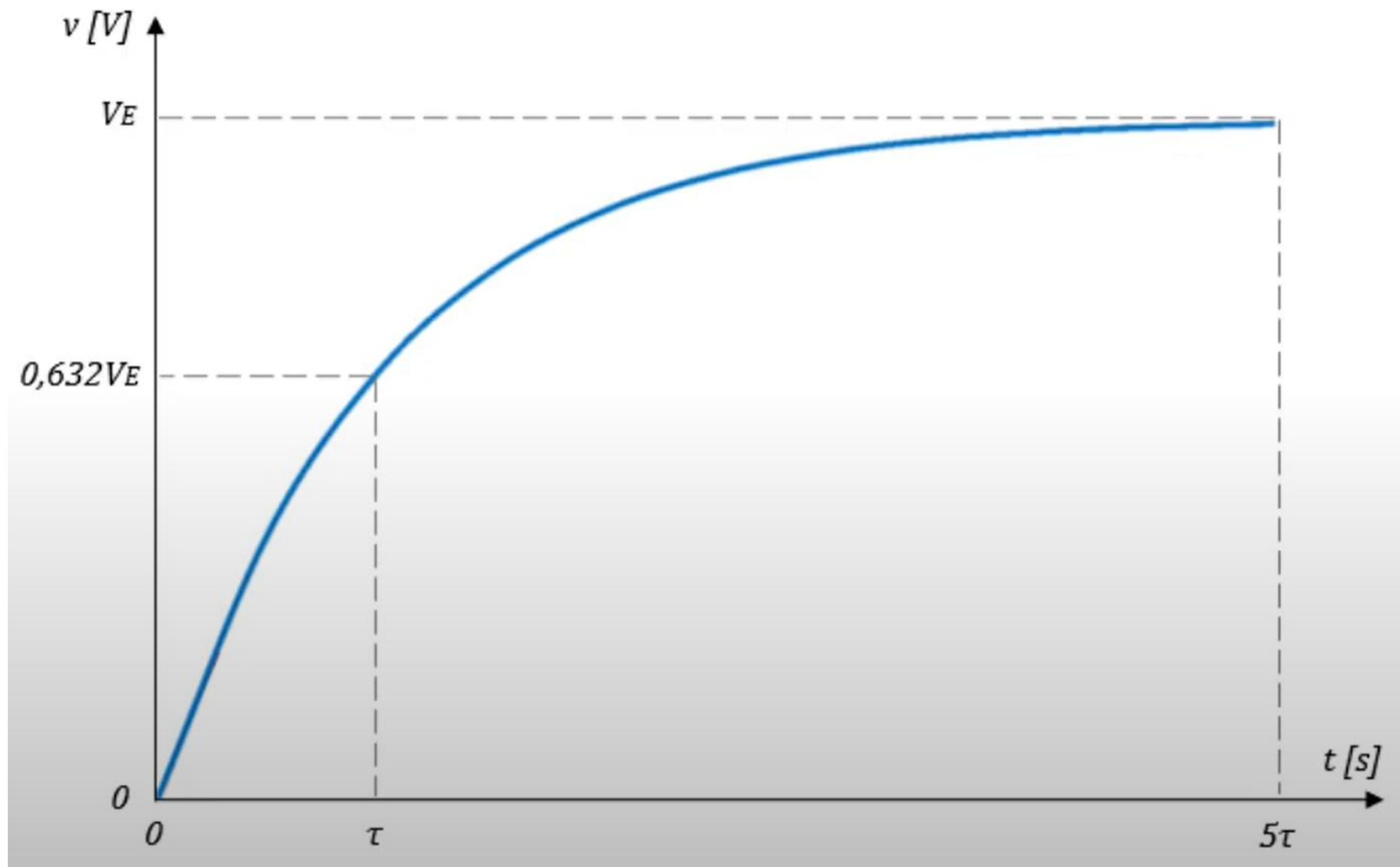
$$v(5\tau) = V_E (1 - e^{-5})$$

$$v(5\tau) = V_E (1 - 0,00674)$$

$$v(5\tau) = 0,993 V_E \approx V_E$$

Carga de un capacitor

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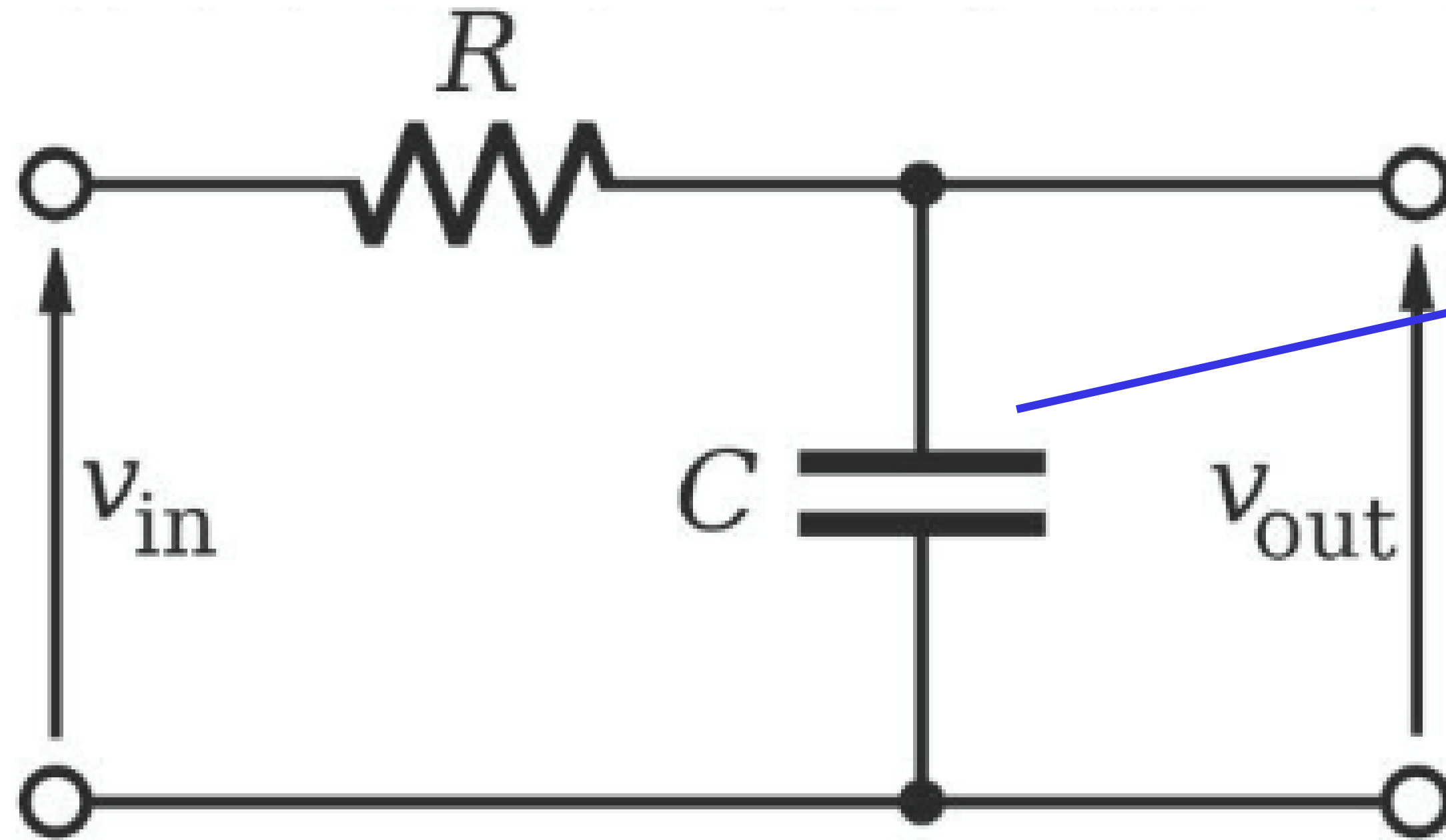


t	$V(t)$
0	0
τ	$0,632 V_E$
5τ	$= V_E$

Filtro pasa bajo

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El capacitor se comporta como una resistencia, cuando la fuente es alterna AC (Reactancia):



$$X_C = \frac{1}{2 \pi f C}$$

y esta depende de la frecuencia f de la fuente

Se puede definir una frecuencia de corte:

$$f_c = \frac{1}{2 \pi R C}$$

- Para $f < f_c$: el capacitor casi no afecta $\rightarrow$ la señal pasa.
- Para $f > f_c$: el capacitor "cortocircuita" las altas frecuencias a tierra $\rightarrow$ la señal se reduce.

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Conclusions

- Signal types and transducers were introduced and discussed.
- Signal representation in time and frequency domains was analyzed.
- Sampling, quantization, and ADC concepts were covered.
- The Fourier Transform was introduced as a key analysis tool.
- Capacitors and low-pass filters were applied in the lab.

4. Signal Representation

Internet of Things